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Realized volatility forecasting for new issues and spin-offs using multi-source transfer learning

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ABSTRACT

Forecasting the volatility of financial assets is essential for various financial applications. This paper addresses the challenging task of forecasting the volatility of financial assets with limited historical data, such as new issues or spin-offs, by proposing a multi-source transfer learning approach. Specifically, we exploit complementary source data of assets with a substantial historical data record by selecting source time series instances that are most similar to the limited target data of the new issue/spin-off. Based on these instances and the target data, we estimate linear and non-linear realized volatility models and compare their forecasting performance to forecasts of models trained exclusively on the target data, and models trained on the entire source and target data. The results show that our transfer learning approach outperforms the alternative models and that the integration of complementary data is also beneficial immediately after the initial trading day of the new issue/spin-off.

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1. Introduction

Volatility forecasting is of significant importance in financial market trading and risk management. In the past, a range of methods for predicting future volatility has emerged, starting with the stochastic volatility models of Taylor (1982), and the ARCH and GARCH models of Engle (1982) and Bollerslev (1986), respectively, as well as extensions thereof. The availability of high-frequency financial data has led to the development of new volatility measures like realized volatility (Andersen et al., 2001) and corresponding forecasting models that aim at modeling realized volatility, such as the popular heterogeneous autoregressive (HAR) model proposed by Corsi (2009). The linear HAR model captures the persistence of realized volatility by incorporating daily, weekly, and monthly components. Its combination of simplicity and strong predictive performance has led to its widespread adoption and further extensions.

Many of these developments build on the HAR framework by incorporating additional predictors of financial volatility. For example, Andersen et al. (2007) decompose volatility into continuous and jump components to account for the distinct persistence of shocks in the jump and continuous volatility process (HAR-C and HAR-CJ). To address the leverage effect, Corsi and Renò (2012b) introduce past aggregated negative returns (LHAR-CJ), while Patton and Sheppard (2015) employ semivariance as an alternative measure. Other variants augment the HAR model with expectations of future volatility, such as implied volatility, or with macroeconomic predictors (Christensen et al., 2023). Additionally, factor-augmented extensions consider cross-market factors, such as those derived from global equity markets (Kim & Baek, 2019; Zhang et al., 2020), to improve forecast accuracy.

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Another line of research examines the lag structure implicitly imposed by the HAR model. Considering a flexible lag structure, Audrino and Knaus (2015) and Audrino et al. (2019) apply (adaptive) LASSO methods and show that, although the HAR-implied lag structure cannot be fully recovered, comparable forecasting performance can be achieved. Building on this approach, Yao et al. (2019) demonstrate that employing a clustered group LASSO improves forecast accuracy. Moreover, LASSO and other data-driven regularization techniques have been applied to HAR models with extended predictor sets, as in Christensen et al. (2023).

Further HAR model extensions focus on measurement errors and/or non-linearities in realized volatility, such as time-varying realized volatility (Corsi et al., 2008) and time-varying coefficients (Chen et al., 2010). For instance, Corsi et al. (2008) introduce the HAR-GARCH model, McAleer and Medeiros (2008) propose a multiple-regime smooth transition HAR (HARST) model, and Wang et al. (2017) examine HAR models with time-varying coefficients. To address measurement errors more explicitly, Bollerslev et al. (2016) introduce the HARQ model, which adjusts model parameters in proportion to the degree of measurement error estimated via realized quarticity. Similarly, Buccheri and Corsi (2021) propose the score-HAR-Kalman (SHARK) model to account for both measurement errors and non-linearities.

More recently, attention has shifted toward machine learning approaches for realized volatility forecasting. These methods are particularly effective at capturing non-linear dynamics. Artificial neural networks (ANNs), for example, can model complex, non-linear relationships without requiring strong parametric assumptions. In this context, Hillebrand and Medeiros (2010) propose a neural network HAR variant, while Pradeepkumar and Ravi (2017) consider particle swarm-optimized quantile regression neural networks. Bucci (2020) evaluates various ANN architectures, including recurrent neural networks such as long short-term memory (LSTM) networks and non-linear autoregressive (NAR) neural networks. Several studies combine LSTMs with convolutional neural networks (CNNs) for volatility forecasting, such as A. and Kristjanpoller (2020) and Zhang et al. (2025), or consider CNNs integrated with alternative model architectures, like Hu et al. (2025) and Souto (2025a). Dilated convolutional networks have been explored by Reisenhofer et al. (2022) and Moreno-Pino and Zohren (2024), while newer architectures such as neural basis expansion analysis (NBEATS; Souto & Moradi, 2024), neural hierarchical interpolation for time series forecasting (NHITS; Souto, 2025b), and transformer-based models (Lim et al., 2025) have also been adopted for volatility forecasting.

Beyond neural networks, tree-based ensemble methods have become widely adopted for realized volatility forecasting. Random forest models are applied by Luong and Dokuchaev (2018), while gradient boosting techniques are examined by Mittnik et al. (2015) and Branko et al. (2024). Extreme gradient boosting (XGBoost) has been employed for realized volatility forecasting across a range of S&P 500 assets and forecast horizons by Teller et al. (2022). Finally, comprehensive comparative studies, such as those by Christensen et al. (2023) and the more recent one by Li and Tang (2025), provide systematic evaluations of multiple machine learning methods. The literature on realized volatility modeling is extensive, and this review is necessarily selective. Nonetheless, it highlights the substantial efforts dedicated to developing models that yield accurate forecasts of realized volatility.

Notably, previous realized volatility forecasting studies, including the aforementioned studies, focus on financial assets with extensive historical data. This is unsurprising, as even parsimonious forecasting methods such as the HAR model require adequate data set sizes to reproduce volatility persistence and generate accurate forecasts. However, this also means that the literature has not yet considered realized volatility forecasts for financial assets with very limited historical data, i.e. financial new issues and spinoffs. It can be assumed that the main reason for this shortage in the existing research is the lack of historical data at the beginning of a financial asset's life cycle. In this paper, we propose to overcome this limitation through complementary data from financial assets with extensive historical records in combination with a transfer learning framework.

Transfer learning represents a common knowledge transfer approach in fields such as natural language processing (Blitzer et al., 2006; Houlsby et al., 2019), computer vision (Deng et al., 2018; Hussain et al., 2019), medical applications (Goetz et al., 2016; Lan et al., 2019), and in industrial settings (Zhou et al., 2023). In financial contexts, transfer learning is considered by Zhang et al. (2023), for example, who forecast financial return series using an adversarial domain adaptation model, and by He et al. (2023), who predict stock price movements using a transfer learning attention mechanism. Furthermore, He et al. (2020) propose multisource transfer learning methods to forecast one-day-ahead stock prices.

However, the existing literature has largely ignored transfer learning for volatility forecasting in general, and for predicting the future volatility of new issues or spinoffs in particular. Although Hao et al. (2025) present an early parameter-based transfer learning application by fine-tuning a hybrid GARCH–GRU model to predict carbon price volatility, their approach neither extends to realized volatility measures nor considers highly data-scarce settings. This limitation in addressing data-scarce environments likely reflects the need for sufficient target domain data during fine-tuning to ensure reliable model adaptation and control overfitting. Other studies have primarily focused on developing global volatility models and applying them to assets excluded from the model's training data sets, or on reusing optimized hyperparameters across forecasting models—approaches that, while related, do not fully capture the principles of transfer learning as formally defined in the foundational transfer learning literature (e.g. Pan & Yang, 2010).

For instance, Liu et al. (2024) develop a global volatility model trained on pooled data from multiple assets, which outperforms models trained solely on individual assets. Similarly, Moreno-Pino and Zohren (2024) propose DeepVol, a deep neural network based on dilated causal convolutions that extracts predictive information directly from

high-frequency intraday returns rather than aggregated daily, weekly, or monthly components. In their framework, the optimal intraday sampling frequency and conditioning period are selected via a global estimation using all NASDAQ-100 stocks. Thereafter, the model is trained and validated on the first half of the stocks and tested on the second half without re-training or re-validation. This approach yields predictive accuracy superior to that of GARCH and HAR models. In addition, Li and Tang (2025) investigate hyperparameter transfer for realized volatility forecasting by optimizing the hyperparameters of random forests and gradient boosting models on S&P 100 stocks and applying the identified optimal configurations to S&P 500 stocks without re-validation. Although all of these studies demonstrate scalability and predictive improvements, none examines assets with limited historical data, such as new issues or spinoffs. Notably, Moreno-Pino and Zohren (2024) suggest that DeepVol may be advantageous in such data-scarce settings, but no corresponding empirical analysis is provided.

To the best of our knowledge, this study is the first to address the research gap in volatility forecasting for new issues and spinoffs, and more generally, the application of transfer learning to realized volatility forecasting. We propose an instance-based multisource transfer learning approach that selects subsequences from complementary financial assets based on the dynamic time warping (DTW) distance between the new issue/spinoff and complementary data series. The selected subsequences are then combined with all the available target data for model training. Unlike fine-tuning-based transfer learning methods, as considered by Hao et al. (2025), which adapt model parameters and therefore require more substantial target domain data to avoid overfitting and ensure stable convergence, our instance-based approach transfers information at the data level. This framework reduces the amount of target data needed, making it particularly well suited for data-scarce settings such as new issues and spinoffs. Moreover, global modeling approaches often suffer from negative transfer due to dissimilarities between source and target data (Wang et al., 2019). Our framework mitigates this risk by restricting transfers to only the most similar instances. Finally, in contrast to fine-tuning methods, our transfer learning approach is inherently model-agnostic, meaning it can be integrated with virtually any forecasting model for realized volatility. This flexibility allows practitioners and researchers to exploit the diverse range of realized volatility models reviewed in this section, or other suitable models, without restricting the underlying methodological choice.

To examine the effectiveness of instance-based multisource transfer learning, this paper evaluates the performance of well-established and commonly used linear and non-linear realized volatility forecasting models in the novel context of forecasting volatility for new issues and spinoffs. Specifically, we integrate transfer learning with HAR models, feedforward neural networks (FNNs), and XGBoost models. In addition, we compare our transfer learning approach to a naive pooling method that combines the new issue/spinoff target data with the entire complementary data set, thereby effectively estimating global models. The method is structurally similar to that of Moreno-Pino and Zohren (2024), but incorporates target data as in Liu et al. (2024). We also benchmark our transfer learning framework against forecasting models solely estimated on the limited historical data of the new issues/spinoffs. Each model and training data set combination is evaluated for one-day-ahead realized variance forecasts on 10 new issues and spinoffs from 50 to 550 days post-distribution date.

Our analysis is further extended by comparing the proposed transfer learning approach against computationally less intensive HAR extensions to assess whether the added complexity and computational demands of multisource transfer learning are justified. Moreover, to gain deeper insights into the behavior of our instance-selection method, we examine the properties of the subsequences selected by our transfer learning framework. Finally, we conduct a second forecast assessment to evaluate the forecast accuracy immediately after the first trading day of the new issue/spinoff. This includes the most extreme and practically interesting scenario: forecasting realized volatility after only one day of trading. Overall, our findings demonstrate that multisource transfer learning is well suited for forecasting the realized volatility of new issues and spinoffs, outperforming the alternative models and approaches considered.

The remainder of this paper is organized as follows. Section 2 provides an introduction to transfer learning and introduces our novel instance-based transfer learning approach for forecasting volatility. Section 3 presents our data sets, the construction of the realized variance series, and the predictors we employ in this paper. The forecasting models considered (HAR models, FNNs, and XGBoost models) are detailed in Section 4. Section 5 presents the empirical setup, including data preparation steps and model estimation. The empirical results of our forecast evaluation study are reported in Section 6. Section 7 concludes.

2. Transfer learning for volatility forecasting

This section establishes a methodology for leveraging information from financial assets with an extensive data history to improve daily volatility forecasts for data-scarce new issues and spinoffs, even immediately after their first trading day. Before introducing our proposed transfer learning approach in Section 2.3, we briefly review transfer learning in general, followed by instance selection methods for time series data such as realized volatility.

2.1. Transfer learning

Transfer learning, as a paradigm, addresses the challenges of knowledge transfer across disparate tasks and domains. Following Pan and Yang (2010), a domain is defined as $\mathscr{D} = (\mathscr{X}, P(X))$, where $\mathscr{X}$ denotes the feature space and $P(X)$ its marginal probability distribution, with $X = \{x_1, \ldots, x_n\} \in \mathscr{X}$. A task is expressed as $\mathscr{T} = \{\mathscr{Y}, (\cdot)\}$, where $\mathscr{Y}$ is the label space and $(\cdot)$ is the predictive function learned from training pairs (x_i, y_i) with $x_i \in X$ and $y_i \in \mathscr{Y}$.

The objective of transfer learning is to leverage knowledge from one or more source domain–task pairs ($\mathscr{D}_S, \mathscr{T}_S$), represented by data D_S, to improve learning in a target domain–task pair ($\mathscr{D}_T, \mathscr{T}_T$) with data D_T, particularly in scenarios where $\mathscr{D}_S \neq \mathscr{D}_T$ or $\mathscr{T}_S \neq \mathscr{T}_T$. The source data set D_S and target data set D_T each consist of feature–label pairs pertaining to their respective domains and tasks.

According to Pan and Yang (2010), transfer learning can be classified as deductive ($\mathscr{D}_S \neq \mathscr{D}_T$), where target labels are unavailable, or inductive ($\mathscr{T}_S \neq \mathscr{T}_T$), where at least a limited number of target labels are available to induce the target predictive function. Unsupervised transfer learning refers to scenarios where no labels are available in either the target or source data. Furthermore, transfer learning methods can be categorized by the type of knowledge being transferred, such as parameters, features, relational knowledge, or instances (Pan & Yang, 2010).

Given the limited labeled data availability immediately after the first trading day of new issues/spinoffs, we deliberately disregard parameter-based methods, as they typically require a large number of labeled target observations. Similarly, we do not consider feature-based methods, since they rely on estimating feature mappings based on target observations, which are scarce for new issues and spinoffs. Finally, as relational-knowledge-based methods are effective only in domains with clear relational structures, such as graphs or networks, but are not applicable to time series transfer learning, we focus primarily on instance-based approaches in an inductive time series transfer learning framework. Specifically, we adopt a multisource transfer learning approach to facilitate knowledge transfer from multiple financial assets, i.e. source domains.

2.2. Time series instance and source selection

Instance-based transfer learning for time series focuses on selecting or weighting source series to improve a target learner (Weber et al., 2021). Since weighting methods are primarily applied in ensemble model settings (Weber et al., 2021), we focus exclusively on instance selection. These methods typically rely on a similarity measure, i.e. some notion of distance or divergence, to identify the source series most relevant to the target domain. This similarity measure serves as a proxy for differences between the marginal probability and/or conditional probability distributions of the target and source domain data sets. By including only the most similar instances, instance selection mitigates negative transfer, where unrelated source instances can degrade predictive performance (Wang et al., 2019).

The literature proposes various approaches for evaluating similarity between target and source instances. For example, Vercruyssen et al. (2017) select subsets of source instances based on their affiliation and proximity to target data clusters, or by applying a density-based method. In the context of source selection, which refers to the selection of one or more data sets consisting of multiple time series from different domains, several studies have adopted thresholds on distance and similarity measures to identify relevant source data sets. These distance and similarity measures include the DTW distance (Ismail Fawaz et al., 2018), Jensen–Shannon divergence (Ye & Dai, 2021), Pearson correlation (Xiao et al., 2014), or maximum mean discrepancy (Islam et al., 2019). Although Weber et al. (2021) classify source selection as a preprocessing step for instance selection or other transfer learning methods, these similarity metrics are equally applicable to instance selection itself.

2.3. Transfer learning for forecasting volatility of new issues/spinoffs

Building on the time series instance and source selection methods discussed in the previous section, we propose a multisource transfer learning approach for predicting the one-day-ahead volatility of new issues and spinoffs in the time period after their first trading day.

To outline our method, we define a combined source data set D_S consisting of P financial assets with an extensive data history by

$$D_S = \left\{ D_{S_p} \mid D_{S_p} = \left\{ (x_t^{(D_{S_p})}, y_t^{(D_{S_p})}) \right\}_{t=1}^{N^{(D_{S_p})}}, p = 1, 2, \ldots, P \right\}, \tag{1}$$

where D_{S_p} denotes the data set of the pth financial asset, $x_t^{(D_{S_p})}$ represents the input features for day t of asset p, which include past volatility measures, such as lagged daily volatility, as well as lagged weekly and monthly volatility components, as defined in Section 3.2. Furthermore, $y_t^{(D_{S_p})}$ represents the label that is to be predicted on day t for asset p, i.e. the daily volatility for the following day, $t + 1$; and $N^{(D_{S_p})}$ denotes the total number of daily observations for asset p. In addition, we define the data set of a target asset, i.e. the new issue/spinoff, T by

$$D_T = \{(x_t^{(D_T)}, y_t^{(D_T)})\}_{t=1}^{N^{(D_T)}}, \tag{2}$$

where $y_t^{(D_T)}$ denotes the daily volatility of the target asset for day $t + 1$, and $x_t^{(D_T)}$ represents the input features of the target asset for day t, again consisting of lagged daily volatility as well as lagged volatility components. Note that the specific composition of the input features depends on the forecast horizon. For shorter forecast horizons, particularly those less than one month or one week, certain components may be unavailable and therefore excluded from the input features. For instance, when predicting volatility one day after the first trading day, the input features include only the first lag of daily volatility. $N^{(D_T)}$ represents the number of observations in D_T available for the target asset.

Since our goal is to forecast the volatility of new issues/spinoffs in the period following their first trading day, $N^{(D_T)}$ is typically small, making it difficult to predict future volatility. Therefore, we aim to extend D_T by including additional data from D_S, which is most similar to D_T. This raises two challenges. First, mismatches in the length of the source and target series complicate the application of similarity measures. Second, the

presence of structural breaks in volatility series, as documented in prior research (Choi et al., 2010; Yang & Chen, 2014, among others), can lead to biased similarity measurements for large source asset data sets.

To address these issues, we split the source series into non-overlapping subsequences and selectively transfer those subsequences that show the highest similarity to D_T. The use of subsequences to account for time series of different lengths has also been considered by Xiao et al. (2014). The choice of non-overlapping subsequences over overlapping ones is primarily based on considerations of computational efficiency. Overlapping subsequences necessitate a greater number of subsequent similarity measurements, significantly increasing the computational burden.

For a source asset D_{S_p}, we define the set of non-overlapping subsequences as follows. Let m represent the length of a single subsequence, and let K be the total number of non-overlapping subsequences that can be generated from the data set, where $K = \left\lfloor \frac{N^{(D_{S_p})}-e}{m} \right\rfloor$. Further, let e denote the number of excess observations, calculated as $e = N^{(D_{S_p})} \pmod m$. The corresponding set of non-overlapping subsequences is then given by

$$D_{S_p}^{(u)} = \Big\{ D_{S_{p,k}} \mid D_{S_{p,k}} = \{ z_{t=(e+1)+(k-1)m}^{(D_{S_p})}, z_{t=(e+1)+(k-1)m+1}^{(D_{S_p})}, \ldots, z_{t=(e+1)+km-1}^{(D_{S_p})} \}, k = 1, 2, \ldots, K \Big\}, \tag{3}$$

where $z_{t=(e+1)+(k-1)m}^{(D_{S_p})}$ represents its respective input feature and label pair:

$$(x_{t=(e+1)+(k-1)m}^{(D_{S_p})}, y_{t=(e+1)+(k-1)m}^{(D_{S_p})}). \tag{4}$$

Subsequences generated in this way omit excess observations at the beginning of each source asset data set. The interval length (m) inherently represents a tradeoff, where small values of m can also lead to an increased computational load, due to the number of similarity comparisons required and a decreased robustness with respect to multi-day model re-estimation intervals, while large values of m lead to large subsequences that potentially contain structural breaks. The resulting combined set, containing the subsequences of all elements of D_S, is defined by

$$U = D_{S_1}^{(u)} \cup D_{S_2}^{(u)} \cup \cdots \cup D_{S_P}^{(u)}. \tag{5}$$

We measure similarity between the target asset and each source subsequence in U using the DTW distance, which, unlike the Euclidean distance, allows for non-linear alignments and is less sensitive to temporal shifts or distortions, thereby capturing similarities between time series that may evolve at different rates (Ratanamahatana & Keogh, 2004). However, computing the pairwise DTW distance between N-length sequences requires $O(N^2)$ time, whereas the Euclidean distance can be computed in $O(N)$ time (Bringmann & Kunnemann, 2015). Specifically, the DTW distance is computed between the lagged volatility predictor series (i.e. lagged daily volatility and lagged volatility components) of the target asset and the generated source subsequences. We address structural breaks and align the sequences by considering only the most recent (i.e. the last) m input feature values of D_T. This also includes the input features for the day of the forecast ($x_{N^{(D_T)}+1}^{(D_T)}$), which notably lack a corresponding output label. Although the labels of the target observations and source subsequences are not directly compared, their similarity is implicitly inferred via the lagged daily volatility. Thus, we account not only for the similarity between the marginal probability distributions, but also indirectly for the similarity between the conditional probability distributions. A more detailed explanation of DTW is provided in Appendix A.

Subsequences with DTW distances below a percentile threshold ϵ of the empirical DTW distance distribution are selected and combined with D_T. In effect, ϵ serves as a selection threshold, determining the number of source subsequences to be included. The resulting pooled data set, which contains the entire available target data along with the selected source data, can then be used to train an arbitrary realized variance prediction model. In this paper, we use the resulting data set to estimate the HAR, FNN, and XGBoost models.

In summary, our proposed transfer learning approach leverages historical data from source assets to enhance target asset volatility forecasts when only limited target observations are available. It effectively addresses data scarcity, variable series lengths, and structural breaks while remaining model-agnostic and requiring only minimal target data. Before discussing the models employed in this paper, we first introduce our data sets along with the predictors considered. Detailed information on the implementation of our transfer learning approach is provided in Section 5.

3. Data, realized variance, and predictor sets

In the following, we introduce the target data set containing the new issues/spinoffs and the source data set containing complementary stock data. We also briefly present two sets of predictors that we use to forecast realized variance.

3.1. Data

As our primary target data set, we use one-minute intraday stock price data obtained from FirstRate Data LLC on 10 new issues and spinoffs that entered the public market between 2010 and 2020 and were part of the S&P 500 index on November 1, 2022.[1] We restrict ourselves to S&P 500 constituents to ensure data availability, liquidity, and comparability. To limit sector concentration (recent new issues skew toward the tech sector) and reduce potential sector bias in model evaluation, we include at most one representative per GICS sector (Standard & Poor's, 2023). Stocks from the Utilities sector are not included in

[1] Note that in this paper, a new issue refers to a private firm going public for the first time. This is also known and referred to as an initial public offering (IPO).

Table 1
Overview of new issues and spinoffs considered.

Ticker	First trading day	Type	Name	GICS sector
TWTR	07.11.2013	New issue	Twitter Inc.	Communication Services
NCLH	18.01.2013	New issue	Norwegian Cruise Line Ltd.	Consumer Discretionary
LW	10.11.2016	Spinoff	Lamb Weston Holdings	Consumer Staples
PSX	01.05.2012	Spinoff	Phillips 66	Energy
SYF	31.07.2014	Spinoff	Synchrony Financial	Financials
MRNA	07.12.2018	New issue	Moderna Inc.	Health Care
CARR	03.04.2020	Spinoff	Carrier	Industrials
DXC	03.04.2017	Spinoff	DXC Technology	Information Technology
CTVA	24.05.2019	Spinoff	Corteva Inc.	Materials
INVH	01.02.2017	New issue	Invitation Homes Inc.	Real Estate

our target data set, as none of these stocks started public trading during our observation period. This design balances market coverage with a feasible sample size. Table 1 provides an overview of the new issues and spinoffs considered and their respective GICS sectors.

Furthermore, for our source data set, we consider the one-minute intraday price data of 66 additional S&P 500 stocks. Again, the data were retrieved from FirstRate Data LLC. The assets considered were selected based on their market capitalization within each GICS sector on November 1, 2022. In particular, we select those assets that are ranked in the top two, bottom two, or at the median level of market capitalization among all stocks in the respective GICS sector that began public trading at least one year prior to the new issue or spinoff within the same GICS sector.[2] To ensure a comprehensive representation of each GICS sector in the source data set, we also include six assets from the Utilities sector. Again, these assets reflect a broad range of market capitalization, specifically selecting the top two, bottom two, and median-ranked Utilities assets that began trading publicly prior to the most recent new issue/spinoff.

The source data set covers the period between January 1, 2010, and September 30, 2022,[3] and is used as complementary data for the naive pooling method and the transfer learning approach of our study. Table B.9 of Appendix B presents a complete list of all stocks in our source data set.

The raw data from both the target and source data set are processed by excluding entries from 4:00 p.m. EST on Friday through 9:30 a.m. EST on Monday, as well as trading activity between 4:00 p.m. and 9:30 a.m. EST for consecutive business days.

[2] If the S&P 500 GICS sectors consist of an odd number of stocks, the median asset and the asset closest to the median in terms of market capitalization are selected. Conversely, if the sectors consist of an even number of stocks, we select the two assets surrounding the median value.

[3] The tickers META, QRVO, and AMT contain a limited number of observations because they began public trading after January 1, 2010. Specifically, META (Communication Services) started trading on May 18, 2012, QRVO (Information Technology) started trading on January 2, 2015, and AMT (Real Estate) entered the public market on March 1, 2012. The same holds true for WRK (Utilities), which began public trading on June 24, 2015, and LIN (Utilities), which started trading on the New York Stock Exchange on November 29, 2018.

3.2. Realized variance and predictor sets

As shown by Andersen et al. (2001), Barndorff-Nielsen and Shephard (2002), and Andersen and Bollerslev (1998), the unobservable daily return variation can, under certain assumptions, be consistently estimated by the sum of squared intraday returns as the sampling frequency (h) of these returns approaches infinity. The corresponding non-parametric estimator, commonly referred to as daily realized variance, is defined by

$$RV_{t,d} = \sum_{j=1}^{h} r_{t,j}^2, \tag{6}$$

where $r_{t,j}$ denotes the continuously compounded jth intraday return sampled at frequency h on day t. Specifically, $r_{t,j}$ is defined as

$$r_{t,j} = \ln(P_{t,j}) - \ln(P_{t,j-1}), \quad j = 1, \ldots, h, \tag{7}$$

where $P_{t,j}$ denotes the asset price at the end of intraday interval j on day t, with $P_{t,0}$ representing the opening price and $P_{t,h}$ the closing price.

Realized variance is typically constructed from five-minute intraday returns ($h = 78$, for standard NYSE and Nasdaq trading hours, 9:30 a.m. to 4:00 p.m. EST), which provides a favorable tradeoff between microstructure noise, e.g. caused by bid–ask bounces or rounding errors, and estimation accuracy (Andersen et al., 2001). We follow the existing literature (Ahoniemi & Lanne, 2013; Christensen et al., 2023, among others) and construct daily realized variance series for all target and source assets from five-minute intraday returns. Our realized variance measure, thus, does not incorporate overnight returns. However, close-to-open price changes can account for a substantial share of total daily volatility; see e.g. Hansen and Lunde (2005). While including overnight returns—typically by adding the squared overnight return to the intraday realized variance (Li & Tang, 2025), or by scaling realized variance (Ahoniemi & Lanne, 2013; Hansen & Lunde, 2005; Martens, 2002)—would yield a volatility proxy that fully reflects close-to-close variation, we deliberately focus on intraday dynamics here.[4] This

[4] Since we consider individual stocks, this is in line with Ahoniemi and Lanne (2013), who find that for individual stocks it is beneficial to exclude the overnight information from the realized variation measure.

allows us to highlight the transfer learning framework under severe data limitations, while leaving the integration of overnight information as a promising extension for future research. Thus, the objective of this paper is to forecast the one-day-ahead realized variance ($RV_{t+1,d}$) of a given target asset.

The related literature considered a variety of different predictors to forecast daily RV; see e.g. Corsi (2009), Patton and Sheppard (2015), Duong and Swanson (2015), Mittnik et al. (2015), Christiansen et al. (2012), or Kambouroudis et al. (2021). In this paper, we primarily use the two sets of predictors recently evaluated by Christensen et al. (2023). They consider a basic predictor set consisting exclusively of the standard HAR volatility components, and an extended predictor set that additionally includes firm-specific and macroeconomic variables. With respect to the more comprehensive predictor set, Christensen et al. (2023) report superior one-day-ahead forecasting accuracy across a variety of different forecasting models compared to a standard HAR model. In the following, we briefly review both of these predictor sets.

The first predictor set, hereafter referred to as Q_{std}, contains the daily ($RV_{t,d}$), weekly, and monthly volatility components of the standard HAR model (Corsi, 2009), where the weekly volatility component is defined by

$$RV_{t,w} = \frac{1}{5}\sum_{i=0}^{4} RV_{t-i,d}, \tag{8}$$

and the monthly volatility component is defined by

$$RV_{t,m} = \frac{1}{22}\sum_{i=0}^{21} RV_{t-i,d}. \tag{9}$$

The second predictor set, referred to as Q_{ext}, additionally includes the one-week price momentum defined by

$$MOM_t = p_{close,t} - p_{close,t-5}, \tag{10}$$

where $p_{close,t}$ denotes the log closing price of the respective asset on day t. It also includes the first-differenced log-transformed dollar trading volume:

$$DV_t = \log(\exp(p_{close,t}) \times v_t) - \log(\exp(p_{close,t-1}) \times v_{t-1}), \tag{11}$$

where v_t is the trading volume on day t. To capture potential volatility shifts due to earnings information, Q_{ext} also includes a binary earnings announcement indicator (EA) that is set to 1 if an earnings announcement is scheduled for the next day and 0 otherwise. Lastly, Q_{ext} contains several macroeconomic indicators: the first-differenced US 3-month Treasury bill rate ($US3M$); the daily squared log return of the Hang Seng stock index (HSI); the Aruoba–Diebold–Scotti business conditions index (ADS; Aruoba et al. (2008)); the economic policy uncertainty index (EPU), introduced by Baker et al. (2016); and the CBOE volatility index (VIX), which captures market expectations regarding the volatility of the S&P 500 index.[5] Notably, our extended data set Q_{ext} coincides with that of Christensen et al. (2023), but excludes the implied volatilities of individual assets derived from options contracts, as options contracts are not available immediately after the initial trading day for most new issues and spinoffs. The final extended predictor set thus comprises 11 variables.

4. Realized variance forecasting models

Our proposed transfer learning approach offers flexibility as it is model-agnostic, making it compatible with any forecasting model. The wide range of models developed to predict realized volatility of assets with extensive historical data sets provides a diverse set of candidate models to consider. From this substantial pool, we focus on well-established models, selecting both linear time series models and non-linear, data-driven models that have demonstrated robust and strong forecasting performance in prior studies. Specifically, we consider linear HAR models (Andersen et al., 2007; Corsi & Renò, 2012a; Patton & Sheppard, 2015), as well as non-linear approaches such as FNNs and XGBoost models (Bucci, 2020; Christensen et al., 2023; Hamid & Iqbal, 2004; Mittnik et al., 2015; Teller et al., 2022).[6] We briefly review these models, with an emphasis on the specific configurations used in this paper.

4.1. HAR model

The HAR model, originally introduced by Corsi (2009), is primarily characterized by its lagged volatility components, which are measured over different temporal horizons, i.e. a day, a week, and a month. This allows the HAR to approximate the slowly decaying (possibly long-memory) autocorrelation often observed in realized variance series. HAR models can be estimated by ordinary least squares (OLS), making them both straightforward to implement and computationally efficient. In its standard form, the HAR model is defined by

$$RV_{t+1,d} = \beta_0 + \beta_d RV_{t,d} + \beta_w RV_{t,w} + \beta_m RV_{t,m} + \epsilon_{t+1,d}. \tag{12}$$

Additionally, we extend the HAR model with the variables from Q_{ext}, as discussed in the previous section, resulting in the following HAR-EXT specification:

$$\begin{aligned} RV_{t+1,d} = {} & \beta_0 + \beta_d\, RV_{t,d} + \beta_w\, RV_{t,w} + \beta_m\, RV_{t,m} + \beta_{mom}\, mom_t \\ & + \beta_{DV}\, DV_t + \beta_{EA}\, EA_t + \beta_{US3M}\, US3M_t + \beta_{HSI}\, HSI_t \\ & + \beta_{ADS}\, ADS_t + \beta_{EPU}\, EPU_t + \beta_{VIX}\, VIX_t + \epsilon_{t+1,d}. \end{aligned} \tag{13}$$

[5] The earnings announcement schedule and HSI data were retrieved from Yahoo Finance. The $US3M$ data are available from Board of Governors of the Federal Reserve System (US) (2023). The ADS data can be accessed in Federal Reserve Bank of Philadelphia (2023), and the EPU data are provided by Economic Policy Uncertainty (2023). The VIX data were retrieved from Cboe Exchange (2023).

[6] Although Christensen et al. (2023) report superior forecast accuracy for random forest models over standard gradient boosting, we adopt XGBoost, which enhances gradient boosting with second-order optimization and additional regularization. Its superior scalability to large data sets relative to random forests (Bentéjac et al., 2021) aligns with the larger training sets generated by our multisource transfer learning framework. Moreover, XGBoost has consistently outperformed random forest models in comparable forecasting contexts (e.g. Hameed et al., 2025), and (Li & Tang, 2025) also find gradient boosting to outperform random forest models.

We consider both the standard HAR based on Q_{std}, in the following referred to as HAR-STD, and its more comprehensive counterpart, the HAR-EXT, which allows for a direct comparison between Q_{std} and Q_{ext}. Due to their popularity and linear model specification, the HAR specifications can be regarded as reference models against which we compare non-linear machine learning models such as FNNs and XGBoost models.

4.2. Feedforward neural networks

Artificial neural networks (ANNs) are widely applied in various fields, e.g. visual recognition (Yan et al., 2015), machine translation (Devlin et al., 2014), and healthcare (Han et al., 2020), due to their non-linear modeling capabilities. Although a variety of specialized ANN frameworks have been introduced, such as convolutional neural networks (LeCun & Bengio, 1995), long short-term memory networks (Hochreiter & Schmidhuber, 1997), and transformer networks (Vaswani et al., 2017), in this article we focus on the most parsimonious class of ANN models, i.e. FNNs. As pointed out by Christensen et al. (2023), FNNs are able to outperform various other data-driven models when it comes to predicting daily realized variance, while offering favorable computational efficiency in contrast to recurrent neural networks such as long short-term memory networks. A detailed overview of FNNs is provided in Appendix C.

For forecasting the volatility of new issues or spinoffs, we consider a three-hidden-layer FNN. The first hidden layer consists of eight, the second layer of four, and the third layer of two artificial neurons, as suggested by Christensen et al. (2023). Each hidden layer contains ReLU activation functions (Agarap, 2019), defined by

$$f(x) = \max(0, x). \tag{14}$$

The resulting FNN architecture is trained based on Q_{std} (FNN-STD) and Q_{ext} (FNN-EXT) via backpropagation combined with ADAM optimization to minimize a mean squared error loss function.

4.3. Extreme gradient boosting

Gradient boosting, as introduced by Freund and Schapire (1997) and Friedman (2001), is characterized by the sequential training of an ensemble of weak learners, where each weak learner, typically a tree learner, is estimated to correct the residuals of the previous one. This idea was later extended by Chen and Guestrin (2016) in the so-called XGBoost framework, which includes additional regularization and more precise weak learner estimation using second-order Taylor expansion. We briefly outline the architecture and estimation of XGBoost models in Appendix D.

In our empirical application, we estimate XGBoost models based on the mean squared error loss function with 40 boosting iterations, i.e. regression trees, and a maximum tree depth of five, to predict future realized variance. The considered XGBoost models are, like the previously introduced forecasting methods, estimated and evaluated based on the standard HAR predictors in Q_{std} (XGB-STD) as well as on the extended predictor set Q_{ext} (XGB-EXT).

All of the presented models are combined with our transfer learning approach introduced in Section 2.3.

5. Setup of the empirical study

This section presents the partitioning of the data sets into training and evaluation sets and introduces three different forecasting approaches for predicting the volatility of new issues and spinoffs. Furthermore, we detail the estimation of the forecasting models considered, as well as the evaluation metrics and statistical tests used to examine the validity of our results.

5.1. Specification of training and evaluation data sets

We trim a copy of each new issue/spinoff data set after the first 150, 250, 350, 450, and the first 550 trading days. The last 100 trading days of each of the five trimmed data sets are designated as the out-of-sample evaluation set (rolling test set), in which all realized variance forecasting models are re-estimated every five trading days. The preceding observations are used exclusively as training data. To prevent information spillover from future source data observations, we align the end date of each source data set with that of the corresponding new issue/spinoff training data by trimming it accordingly.

Based on this setup, we assess the one-day-ahead realized variance forecasting performance over the entire evaluation set, i.e. over the observations 51 to 550, as well as over the individual 100-day subperiods following the first 50, 150, 250, 350, and 450 trading days of the respective new issue or spinoff. We refer to these sample periods as s, where $s = s^*$ represents the entire 500-day evaluation period, $s = 50$ represents the 100-day evaluation period following the initial 50 trading days, $s = 150$ represents the 100-day evaluation period following the initial 150 trading days, and so forth. Including s^* provides a comprehensive benchmark period that supports a direct, pairwise evaluation of model performance, ensuring a concise and consistent presentation of results. The division of training and evaluation periods for each sample period is illustrated in Fig. 1. It is important to note that, in the target training data sets, the actual number of observations is smaller than the number of available past trading days, due to the inclusion of predictors that are constructed using lagged data, such as weekly or monthly realized variance components. Recall that the monthly realized variance component at time period t is based on the realized variance of day t and the previous 21 daily realized variance observations. Consequently, after 50 trading days, for example, the target training set for a new issue/spinoff consists of only 29 data points.

In addition to evaluating the forecasting performance subsequent to the initial 50 trading days, we also consider the extreme case where forecasts are computed directly after the first trading day of the new issue or spinoff, using correspondingly reduced predictor sets. The approach is described in detail in Section 6.4.

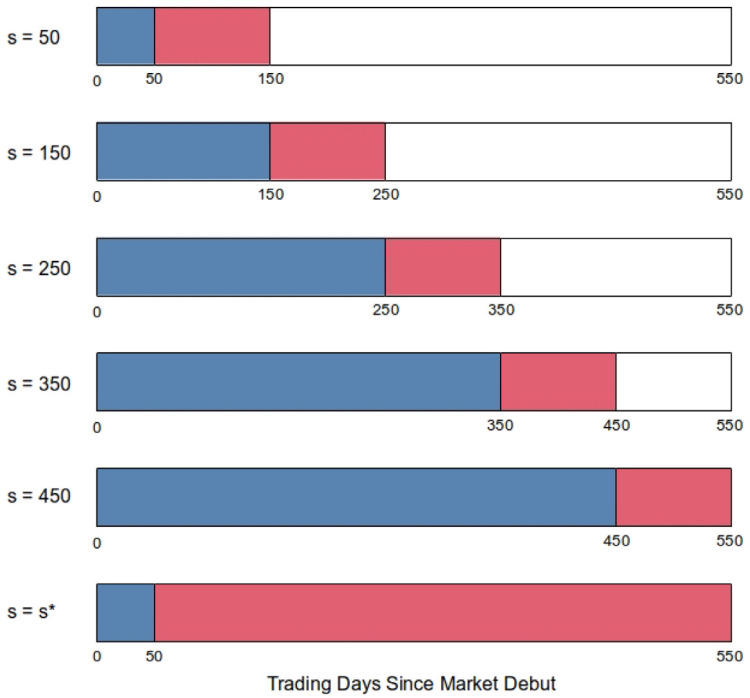


Fig. 1. Training and evaluation data sets for new issues and spinoffs across different sample periods s. Blue intervals represent exclusive training periods, while red intervals indicate evaluation periods during which all realized variance forecasting models are re-estimated every five trading days. An overview of the training and evaluation periods used in our forecast evaluation starting immediately after the first trading day is provided in Fig. 3.

Furthermore, note that the input features for all FNN models are normalized, whereas scaling is omitted for the HAR and XGBoost models, as these models are invariant to scaled input features.

5.2. Forecasting approaches

We consider three forecasting approaches that differ in terms of the training data set used to estimate the various realized variance forecasting models. In the first approach, all forecasting models are estimated exclusively based on the individual target training data of each new issue/spinoff. We refer to these models as target-only (TO) models. In the second approach, the forecasting models are trained on the pooled target training data consisting of the training data of the individual new issue/spinoff and of the entire source data set. The training data sets of these models maintain the temporal alignment between the target and source asset data series, as discussed in the previous section. We refer to models estimated in this manner as naive pooling (NP) models. In the third approach, the realized variance forecasting models are estimated based on the pooled target training data consisting of the training data of the individual new issue/spinoff and of all source subsequences selected by our proposed transfer learning method discussed in Section 2.3. Recall that the number of selected source subsequences is determined by the threshold parameter ϵ. To evaluate its impact, we analyze different inclusion rates (25th, 50th, and 75th percentiles) for ϵ. This allows us to assess how the number of selected subsequences influences the precision of realized variance forecasts. The resulting multisource transfer learning (MTL) models are hereafter referred to as the MTL-25, MTL-50, and MTL-75 models. Finally, we consider a random walk (RW) forecast, which is defined as the previous day's realized variance of the target asset.

5.3. Model estimation

The estimation of the predictive models under consideration is conducted individually for each target asset, predictor set, and training data set approach based on the specific estimation method pertinent to each model class. Specifically, the HAR models are estimated using ordinary least squares (OLS) regression, the XGBoost models are trained using Newton boosting, and the FNN models are optimized using the ADAM optimization algorithm in conjunction with the backpropagation method.

At the beginning of each 100-day sample period, the TO models are estimated based on all available target observations, whereas the NP model training data sets additionally include all source data observations up to the first day of the 100-day sample period. The MTL models are estimated based on the target data and selected source data subsequences. An illustration of the MTL subsequence selection process is provided in Fig. 2. The source subsequences are selected based on their

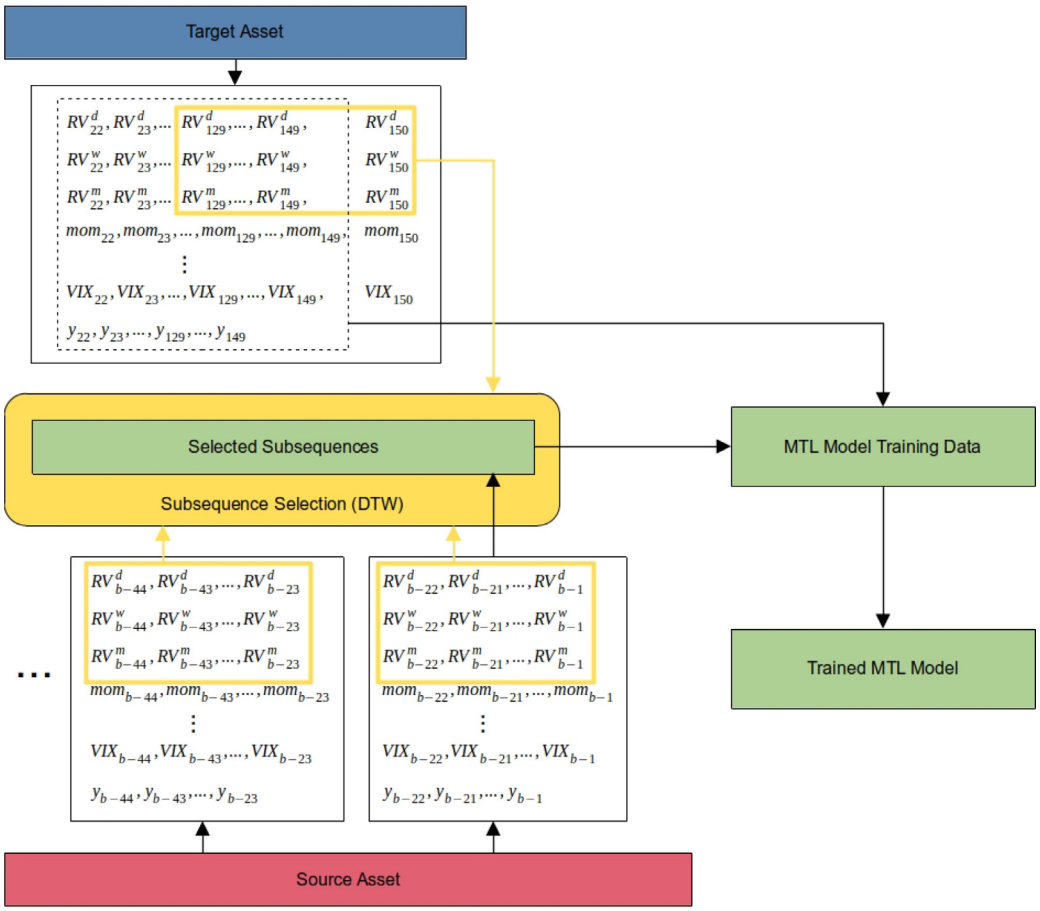


Fig. 2. Example MTL model subsequence selection process after the 150th trading day of a new issue/spinoff. Depicted is the selection of subsequences from a single source asset for a subsequence size of $m = 22$. The date of the source data observation b corresponds to the 150th trading day of the new issue/spinoff. The DTW distance is computed based on the realized variance components. The number of selected subsequences is determined by the threshold parameter ϵ; in the depicted example, only the most recent source subsequence is selected. This process is repeated at each estimation step for the progressively extended target and source series while maintaining temporal alignment.

similarity to the input feature values of the target training data set over the most recent month ($m = 22$), defined as the last 22 trading days. To determine the similarity between the target observations and the source subsequences, we calculate the DTW distance based on the lagged volatility predictors, i.e. RV_d, RV_w, RV_m.[7] Finally, the selected source subsequences are combined with the entire corresponding target training data set.

We then compute one-day-ahead realized variance forecasts, where each model is re-estimated after five consecutive trading days. Consequently, the training data set of the TO models increases by five observations at each re-estimation point. Similarly, the combined training data set of the NP models is expanded by five observations from each source asset and by five observations from the target asset. Subsequences for MTL models are selected based on the current last $m = 22$ target input feature values from the re-estimation point. This approach inevitably means that the data sets of MTL models are also incrementally expanded and continually updated based on the latest target input feature values at each estimation point.

For the FNN models, the training is limited to a maximum of 500 epochs at the first estimation point of each sample period, while an early termination mechanism stops the training prematurely if the mean squared error on a 10% validation set does not improve for 100 epochs. The resulting epoch count is then applied consistently to all subsequent re-estimation points of the same sample period. Unlike NP FNNs, which are trained with a batch size of 1024, and MTL FNNs, which use a batch size of 512, TO FNNs require a significantly reduced batch size of four. This adjustment is attributed to the limited volume of target training data available for new issues/spinoffs. Table 2 presents the adopted hyperparameter values of the FNN and XGBoost models.

Apart from optimizing the epoch count of the FNNs, we do not perform any other hyperparameter optimizations. The reasons for this are twofold. First, the target data are already sparse. Thus, additional splitting to generate validation data would result in extremely small validation

[7] The DTW distance calculations are implemented using the 'tslearn' Python package, as introduced by Tavenard et al. (2020)

Table 2
Hyperparameters of the XGBoost and FNN models.

XGBoost hyperparameters	Value	FNN hyperparameters	Value
No. of tree learners	40	Learning rate	0.001
Max. tree depth	5	No. of hidden layers	3
Shrinkage (learning rate)	0.1	Units per hidden layer	[8,4,2]
No. of leaves regularization (γ)	0.1	Batch size	4/512/1024
L1 regularization	0	Hidden layers act. functions	ReLU
L2 regularization	1	Opt. algorithm	ADAM
Subsample ratio per tree	0.75		
Min. no. of instances per Leaf	1		

sets with questionable robustness regarding validation set errors. Second, the proposed rolling window forecasting and re-estimation methodology is computationally expensive. Additional hyperparameter tuning would significantly increase the computational time. However, XGBoost models and FNNs naturally exhibit a large number of potential hyperparameters. Therefore, it can be expected that these model classes would benefit the most from potential hyperparameter optimizations compared to HAR models.

5.4. Forecast evaluation

Before evaluating the generated realized variance forecasts, we enforce a non-negativity constraint on the predicted values. In particular, we follow Christensen et al. (2023) and replace negative forecast values with the empirical minimum realized variance of the target training data set. Notably, negative forecast values are extremely rare in our application. The out-of-sample performance of the different models is evaluated based on the same metric as is used in the estimation of the FNN and XGBoost forecasting models, i.e. the mean squared error (MSE). In addition, we report the quasi-likelihood loss (QLIKE), which is less sensitive to extreme observations and robust to noise in the volatility proxy (Patton, 2011).

Based on these error metrics, we assess the predictive accuracy of the forecasting models in two ways. First, we analyze the performance of each model over the entire test period, i.e. for $s = s^*$, in a pairwise comparison. To this end, we compute for each new issue/spinoff the MSE (QLIKE) of one model (M_i) relative to the MSE (QLIKE) of another model (M_j) and report the cross-sectional average over all new issues/spinoffs of these pairwise relative MSEs (QLIKE losses). For each target asset, we also examine the significance of the differences in the forecasts of individual model pairs (M_i, M_j) by applying the one-sided Diebold–Mariano (DM) test introduced in Diebold and Mariano (1995), at a 5% significance level. Specifically, we report whether the null hypothesis of equal forecast performance between forecasting models M_i and M_j, i.e. $H_0 : \text{MSE}_i = \text{MSE}_j$ ($H_0 : \text{QLIKE}_i = \text{QLIKE}_j$), is rejected in favor of the alternative hypothesis, $H_1 : \text{MSE}_i < \text{MSE}_j$ ($H_1 : \text{QLIKE}_i < \text{QLIKE}_j$), for more than 50% of the target assets considered.

In a second forecast evaluation, we investigate the forecast accuracy for the individual 100-day sample periods. As this results in large volumes of forecast evaluation measurements, we only report the cross-sectional average of the MSE (QLIKE) of each model relative to the MSE (QLIKE) of the TO HAR-STD model.

Finally, for each sample period and new issue/spinoff, we construct a model confidence set (MCS; Hansen et al., 2011) using a 95% confidence level, which contains the most accurate forecasting models for the MSE (QLIKE) metrics. To account for serial dependence in forecast errors, we employ the stationary bootstrap of Politis and Romano (1994) with 5000 replications, and implement the semi-quadratic test statistic, as described by Hansen et al. (2011), within the MCS framework. We enumerate the instances in which a particular forecasting model is included in the MCS for a new issue/spinoff during a specific 100-day sample period, and present the total number of occurrences for individual 100-day sample periods as well as the cumulative occurrences across all 100-day sample periods. In addition, we generate MCSs for the entire test period $s = s^*$ and report the total number of occurrences of individual forecasting models across all new issues and spinoffs.

6. Results

This section presents our empirical results. We first focus on evaluating the forecasting performance of our multisource transfer learning approach compared to forecasts that are exclusively based on the target data (TO) or based on the entire source and target data (NP). We then discuss in more detail the properties of the source subsequences selected by our transfer learning approach. In addition, we compare the proposed transfer learning approach against computationally less intensive HAR extensions. Finally, we assess the accuracy of our forecasting approaches under the extreme scenario of forecasting immediately one day after the distribution date of each new issue/spinoff.

6.1. Realized variance forecasting for new issues and spinoffs

The evaluation results of the realized variance forecasts, conducted through a pairwise comparison of all considered models across all new issues and spinoffs for the entire test period ($s = s^*$), are presented in Tables E.10 and E.11 in Appendix E for the MSE and QLIKE metrics, respectively. This analysis shows significant differences in the performance of the various models. In particular, the

MTL-75 XGB-EXT model stands out by achieving cross-sectional MSE and QLIKE ratios of one or less relative to all other models.[8] As indicated by the rejection of the null hypothesis of the DM test for more than 50% of all new issues/spinoffs, the MTL-75 XGB-EXT also significantly outperforms the RW and each TO model with respect to both MSE and QLIKE.

Focusing on the aggregated MSE (QLIKE) ratios of each model relative to the TO HAR-STD, i.e. the well-established standard HAR model, it becomes evident that the MTL-75 XGB-EXT leads to the greatest improvement in forecast accuracy, with reductions of 15.4% in MSE and 18.1% in QLIKE. When comparing each model with the RW, the MTL-75 FNN-EXT reduces the MSE by 33.2%, while the MTL-75 XGB-EXT provides a 30.9% reduction in the average QLIKE relative to the RW.

This strong predictive performance of our multisource transfer learning approach relative to the RW and TO HAR-STD is not limited to the aforementioned models, but generalizes to nearly all MTL-based HAR, FNN, and XGBoost models. The only exception is the MTL-25 XGB-STD under QLIKE, which underperforms the TO HAR-STD. In all other cases, MTL models, whether using the standard HAR predictors in Q_{std} or the extended predictor set in Q_{ext}, yield aggregated MSE and QLIKE ratios below one when evaluated against the RW and TO models. In addition, all MTL models demonstrate significantly higher forecast accuracy than the RW in terms of the MSE and QLIKE for more than 50% of all new issues/spinoffs according to the results of the DM tests. The same holds true for the majority of MTL EXT models for both the MSE and QLIKE relative to TO HAR-STD and all other individual TO models.

A comparison of the MTL and NP specifications reveals that each MTL EXT model provides, on average, more accurate forecasts than both its NP STD and NP EXT counterparts in terms of the MSE and QLIKE. However, in most one-to-one comparisons between the MTL EXT and NP models, the null hypothesis of the DM tests cannot be rejected for more than 50% of all new issues/spinoffs. This indicates that MTL EXT models lead to a significant improvement in realized variance forecast accuracy compared to models trained on the entire target and source data set, but not necessarily for every new issue/spinoff. For the standard predictor set, most MTL models achieve higher accuracy than their NP STD counterparts in terms of the MSE, but their performance under QLIKE is mixed, with several results indicating comparable or inferior accuracy.

The analysis of the MTL STD and MTL EXT models shows that, in most cases, each MTL EXT model exhibits a cross-sectional relative MSE and QLIKE of less than one when compared to the corresponding MTL STD model. In particular, the MTL XGB-EXT models achieve improvements in the average MSE of up to 12% over the MTL XGB-STD models. For MTL models other than MTL XGB, the use of the extended predictor set yields smaller improvements in MSE, typically around 5%. In terms of QLIKE, however, the improvements in forecasting accuracy are more pronounced overall, with the XGB-EXT models achieving reductions of up to 16.3%.

Among the MTL EXT models, the MTL XGB-EXT models generally demonstrate higher prediction accuracy than the MTL HAR-EXT and MTL FNN-EXT models in terms of the MSE and QLIKE, followed by MTL FNN-EXT models, which outperform the MTL HAR-EXT models in the majority of cases. The reduction in MSE (QLIKE) of the respective superior model is mostly within a range of up to 5%.

The results of the pairwise comparison based on the entire test sample period show that our multisource transfer learning approach generally yields superior forecasts in comparison to models trained exclusively on the target data. For models using the extended predictor set, multisource transfer learning also consistently outperforms the naive pooling approach, which combines the entire source and target data. The improved performance of MTL models relative to NP models is also evident when using the standard predictor set and focusing on MSE. However, under QLIKE, the results are mixed. This suggests that the benefits of instance selection for forecasting the realized variance of new issues and spinoffs are most pronounced when the predictor set is sufficiently rich. In order to gain further insights into the relevance of the length of the available target data history, we now turn to an analysis of the individual test sample periods.

Table 3 presents the MSE (QLIKE) of each model relative to that of the TO HAR-STD model, averaged across all new issues and spinoffs, for the individual 100-day evaluation periods conditioned on different initial target data lengths (50, 150, 250, 350, and 450 days). We observe that the MTL models using the extended predictor set also dominate in each individual sample period. Except for MTL-25 XGB-EXT in $s = 250$ under the MSE, each MTL EXT model provides, on average, more accurate realized variance predictions than the TO HAR-STD in every sample period, considering both the MSE and QLIKE. Moreover, it is always an MTL EXT model that provides the greatest improvement in MSE and QLIKE over the TO HAR-STD model in each sample period. In particular, in the first sample period, the MTL EXT models yield improvements in forecast accuracy of up to 24% in the average MSE and up to 28.6% in the average QLIKE. In contrast, the NP models show less pronounced improvements. Although they generally outperform their TO counterparts, their gains in forecast accuracy are typically smaller than those achieved by the corresponding MTL EXT models in terms of the MSE and QLIKE, and smaller than those of corresponding MTL STD models under the MSE. Hence, multisource transfer learning, particularly when combined with the extended predictor set, consistently delivers superior forecasts, regardless of the length of the target asset's historical data.

[8] We also estimated random forest models, as in Christensen et al. (2023), for each forecasting approach and predictor set combination. Although the random forest TO models outperformed their XGBoost counterparts, XGBoost achieved superior performance in terms of the MSE and QLIKE in the majority of cases involving additional source data (NP, MTL), including the overall best-performing model, MTL-75 XGB-EXT.

Table 3
MSEs and QLIKE losses of each forecasting model relative to the MSE (QLIKE) of the TO HAR-STD model, averaged over all new issues and spinoffs considered. The MSE and QLIKE metrics are obtained for 100 rolling one-day-ahead forecasts based on $s = 50$, 150, 250, 350, and 450 trading days after the distribution day of the new issue/spinoff. Model names reflect both the forecasting approach and predictor set: TO denotes target-only models trained solely on the target data; NP denotes naive pooling, where models are trained on both the target and all available source data; and MTL denotes multisource transfer learning, where the numerical suffix (25, 50, 75) indicates the percentage of subsequences transferred from the source data into the final training data set. The second part of the name indicates the base model class, with HAR referring to the heterogeneous autoregressive model, FNN to feedforward neural networks, and XGB to XGBoost, while STD and EXT refer to the standard HAR predictor set and an extended predictor set, respectively. Blue values indicate, for each sample period (s) and error metric, the best-performing model within each model class, i.e. HAR, FNN, and XGB. The MSE (QLIKE) criteria of the overall best-performing model are highlighted in bold blue. '> 99' represents values that exceed 99.

Model	s = 50		s = 150		s = 250		s = 350		s = 450	
	MSE	QLIKE	MSE	QLIKE	MSE	QLIKE	MSE	QLIKE	MSE	QLIKE
RW	1.365	1.209	1.479	1.190	1.294	1.295	1.476	1.268	1.360	1.062
TO HAR-EXT	> 99	1.931	1.415	3.948	1.808	1.832	1.204	1.058	1.264	1.089
NP HAR-STD	0.975	0.867	1.002	0.890	0.986	0.974	1.038	0.977	1.011	0.900
NP HAR-EXT	0.902	1.133	1.013	1.117	1.176	1.069	1.071	1.012	1.374	1.058
MTL-25 HAR-STD	0.898	0.897	0.952	0.950	0.933	0.980	0.969	1.014	0.964	0.979
MTL-50 HAR-STD	0.894	0.883	0.940	0.939	0.927	0.974	0.971	1.012	0.957	0.964
MTL-75 HAR-STD	0.891	0.871	0.934	0.926	0.924	0.965	0.968	1.002	0.951	0.946
MTL-25 HAR-EXT	0.794	0.760	0.905	0.899	0.880	0.933	0.892	0.897	0.935	0.914
MTL-50 HAR-EXT	0.788	0.750	0.893	0.894	**0.878**	0.933	0.892	0.896	0.933	0.905
MTL-75 HAR-EXT	0.784	0.742	0.880	0.880	0.891	**0.932**	0.884	0.883	0.947	0.889
TO FNN-STD	1.076	1.479	1.058	1.063	1.085	1.079	0.996	1.009	1.036	1.047
TO FNN-EXT	> 99	1.410	0.909	2.015	1.219	1.262	1.014	0.925	1.361	1.104
NP FNN-STD	0.931	0.854	0.969	0.887	0.938	0.944	1.000	0.977	0.968	0.895
NP FNN-EXT	1.001	1.176	1.150	1.237	1.526	1.317	1.508	1.232	1.613	1.237
MTL-25 FNN-STD	0.898	0.895	0.957	0.950	0.934	0.975	0.973	1.021	0.972	0.990
MTL-50 FNN-STD	0.895	0.883	0.944	0.940	0.929	0.973	0.971	1.013	0.959	0.966
MTL-75 FNN-STD	0.892	0.871	0.936	0.927	0.926	0.966	0.967	1.002	0.952	0.949
MTL-25 FNN-EXT	0.810	0.774	0.903	0.880	0.881	0.944	0.896	0.898	0.953	0.923
MTL-50 FNN-EXT	0.796	0.760	0.880	0.865	0.880	0.939	0.888	0.886	0.938	0.906
MTL-75 FNN-EXT	0.782	0.742	0.847	**0.834**	0.895	0.938	0.880	0.869	0.946	0.882
TO XGB-STD	1.549	1.217	1.470	1.618	1.950	1.230	1.277	1.257	1.296	1.133
TO XGB-EXT	1.323	1.093	1.475	1.051	1.623	1.125	1.278	1.018	1.176	0.982
NP XGB-STD	0.914	0.914	0.947	0.951	1.005	0.992	0.982	0.973	0.984	0.947
NP XGB-EXT	0.833	0.824	0.892	0.917	1.017	1.004	0.996	0.937	0.995	0.953
MTL-25 XGB-STD	0.891	0.891	1.005	1.355	1.808	1.015	0.972	1.028	0.992	0.989
MTL-50 XGB-STD	0.889	0.870	0.963	0.953	1.018	0.990	0.967	1.008	0.968	0.955
MTL-75 XGB-STD	0.881	0.858	0.935	0.921	0.981	0.987	0.959	0.987	0.966	0.929
MTL-25 XGB-EXT	0.788	0.753	0.874	0.916	1.010	0.966	0.883	0.902	0.911	0.867
MTL-50 XGB-EXT	0.780	0.734	0.848	0.884	0.956	0.953	**0.858**	0.879	0.909	0.842
MTL-75 XGB-EXT	**0.760**	**0.714**	**0.830**	0.839	0.935	0.945	**0.858**	**0.857**	**0.905**	**0.835**

The higher average forecast accuracy of MTL EXT models is also reflected in the model confidence sets, as shown in Table 4. Specifically, for the combined sample periods ($s = s^*$), the MTL-75 XGB-EXT is included in three MCSs based on MSEs and six MCSs based on the QLIKE. Taking into account the aggregated occurrences in MCSs of individual sample periods, the MTL-75 XGB-EXT is contained overall in eight MCSs based on MSEs and nine MCSs based on the QLIKE, surpassing all other models. In general, considering both the MSE and QLIKE, we can observe that MTL EXT models (i.e. MTL XGB-EXT, MTL FNN-EXT, and MTL HAR-EXT models) are more frequently included in the MCSs than NP and TO models.

In the following, we assess how the number of selected subsequences, determined by the choice of ϵ, influences the forecast accuracy of the MTL models by comparing the forecasting performance of the MTL-25, MTL-50, and MTL-75 models within each model specification (i.e. HAR, FNN, and XGB) with the same set of predictors (i.e. STD or EXT). The results presented in Tables E.10 and E.11 reveal that there are only minor differences in the forecasting performance, with variations typically within a narrow range of less than 5% for both the relative MSE and QLIKE. However, the relative MSE and QLIKE tend to decrease slightly as more source data are integrated, such as in the MTL-75 models. With respect to MTL EXT models, which outperform their NP EXT counterparts under both the MSE and QLIKE, and to most MTL STD models under the MSE, these findings suggest that there are two types of source data observations. On the one hand, there is a relatively small fraction of source observations that significantly improve the volatility forecasts compared to models trained exclusively on the target data. On the other hand, there is a small set of observations that are detrimental to the prediction. Thus, the main benefit of transfer learning in this context is to identify the former and exclude the latter from the training data sets. As our results suggest, this can be achieved by exploiting the similarity between target and source observations.

After the detailed analysis and discussion of the performance of the MTL models, we now proceed to analyze the performance of the non-MTL models. Comparing the forecasting accuracy of the NP models in relation to the RW, see Tables E.10 and E.11, we find that all NP models show

Table 4
Frequency of inclusion of forecasting models in MCSs based on the MSE and QLIKE of all considered new issues and spinoffs for individual sample periods. Each cell reports the number of MCS inclusions for a given model under the MSE and QLIKE (denoted as MSE/QLIKE). The frequency of inclusion of each model over all individual sample periods is aggregated in a separate column ('Agg.'). The column '$s = s^*$' reports the inclusion of individual forecasting models in the MCSs of new issues/spinoffs for the entire test period. Non-zero entries are highlighted in blue. The model with the highest number of inclusions within each error metric and sample period (s), as well as in the aggregate count, is highlighted in bold blue. Model names reflect both the forecasting approach and predictor set: TO denotes target-only models trained solely on the target data; NP denotes naive pooling, where models are trained on both the target and all available source data; and MTL denotes multisource transfer learning, where the numerical suffix (25, 50, 75) indicates the percentage of subsequences transferred from the source data into the final training data set. The second part of the name indicates the base model class, with HAR referring to the heterogeneous autoregressive model, FNN to a feedforward neural network, and XGB to XGBoost, while STD and EXT refer to the standard HAR predictor set and an extended predictor set, respectively.

Model	s = 50	s = 150	s = 250	s = 350	s = 450	Agg.	s = s*
RW	0/0	0/0	0/0	0/0	0/0	0/0	0/0
TO HAR-STD	0/0	2/1	1/1	0/0	0/0	3/2	0/0
TO HAR-EXT	**3**/0	0/0	0/1	**3**/2	0/1	6/4	0/0
NP HAR-STD	0/0	0/**2**	0/0	0/1	0/0	0/3	0/0
NP HAR-EXT	0/0	0/0	0/0	0/0	0/1	0/1	0/0
MTL-25 HAR-STD	0/1	0/0	0/0	0/0	0/0	0/1	0/0
MTL-50 HAR-STD	0/0	0/0	0/0	0/0	0/0	0/0	0/0
MTL-75 HAR-STD	0/0	1/1	0/0	0/0	1/1	2/2	0/0
MTL-25 HAR-EXT	2/1	0/0	0/**2**	2/0	**1**/0	5/3	0/0
MTL-50 HAR-EXT	2/1	0/0	0/0	2/0	**1**/1	5/2	0/0
MTL-75 HAR-EXT	1/0	0/0	1/1	**3**/1	0/0	5/2	1/0
TO FNN-STD	0/0	0/0	0/0	0/0	**1**/0	1/0	0/0
TO FNN-EXT	0/0	**3**/**2**	0/0	2/2	**1**/0	6/4	0/0
NP FNN-STD	0/0	1/0	1/0	0/0	0/0	2/0	1/0
NP FNN-EXT	2/1	0/1	1/0	0/0	0/0	3/2	0/0
MTL-25 FNN-STD	0/0	0/0	0/0	0/0	0/0	0/0	0/0
MTL-50 FNN-STD	0/0	0/0	0/0	0/0	0/0	0/0	0/0
MTL-75 FNN-STD	0/0	0/0	0/0	0/0	**1**/0	1/0	0/0
MTL-25 FNN-EXT	1/0	1/0	**3**/0	1/0	0/0	6/0	0/0
MTL-50 FNN-EXT	1/0	1/0	0/0	2/0	0/0	4/0	1/0
MTL-75 FNN-EXT	2/1	**3**/1	0/1	1/0	0/0	6/3	2/2
TO XGB-STD	0/0	0/0	0/1	0/0	0/0	0/1	0/0
TO XGB-EXT	0/0	1/0	0/1	0/0	**1**/**2**	2/3	1/1
NP XGB-STD	0/0	2/0	1/1	1/0	**1**/1	5/2	1/0
NP XGB-EXT	0/1	1/0	0/0	1/0	**1**/1	3/2	1/0
MTL-25 XGB-STD	0/0	1/0	0/0	0/0	**1**/0	2/0	0/0
MTL-50 XGB-STD	0/0	1/0	1/0	0/0	**1**/1	3/1	0/0
MTL-75 XGB-STD	1/0	1/0	0/0	0/0	**1**/1	3/1	0/0
MTL-25 XGB-EXT	1/0	1/0	0/1	**3**/0	**1**/0	6/1	0/0
MTL-50 XGB-EXT	1/0	1/0	0/0	**3**/1	**1**/0	6/1	1/1
MTL-75 XGB-EXT	**3**/**4**	1/**2**	1/0	**3**/**3**	0/0	**8**/**9**	**3**/**6**

average MSE (QLIKE) ratios of less than one. At the same time, the majority of the NP models achieve a significant reduction in the MSE and QLIKE loss compared to the RW for more than 50% of the new issues/spinoffs according to the DM tests. Moreover, we observe that specifically the NP HAR and NP XGB models exhibit greater forecast accuracy on average than their TO counterparts when employing the same predictor set. For example, NP XGB-EXT reduces the MSE by 25.1% in relation to TO XGB-EXT. On the other hand, the NP FNN-EXT performs worse than TO FNN-EXT with respect to MSE (QLIKE) ratios. Interestingly, the NP HAR-STD and NP FNN-STD models outperform their EXT counterparts when compared to each other, while the NP XGB-EXT exhibits small improvements in the MSE and QLIKE loss, when compared to the NP XGB-STD model. Contrary to these observations, the NP HAR models are generally less frequently included in the model confidence sets of individual sample periods than the TO HAR models, see Table 4. Meanwhile, the NP XGB models are included more frequently in the model confidence sets of individual sample periods than the TO XGB models. In summary, models trained on both the target and entire source data (i.e. NP models) generally outperform both the RW and their TO counterparts in terms of forecast accuracy, with the exception of NP FNN-EXT. We also note that less complex NP STD models, such as NP HAR-STD and NP FNN-STD, outperform their more complex EXT counterparts, suggesting that extending the predictor set is not necessarily beneficial in a naive pooling framework without transfer learning.

Concerning TO models, our results show that the TO HAR-STD model consistently outperforms the RW and all other TO models in terms of average MSE (QLIKE) ratios. Furthermore, we observe that the TO HAR-STD and the TO FNN-STD demonstrate superior performance compared to the RW and their TO EXT counterparts. When considering the analysis of individual sample periods in Table 3, this distinction appears to be most pronounced during the initial sample period, where the TO HAR-EXT and TO FNN-EXT models yield substantial MSEs and QLIKE losses relative to the TO HAR-STD. These findings are consistent with the general understanding that parsimonious models

tend to excel when trained on limited data, as opposed to more complex models. Interestingly, this contrast is less pronounced when considering TO XGB-EXT, which outperforms TO XGB-STD in terms of the MSE and QLIKE. However, both the TO XGB-STD and TO XGB-EXT models simultaneously perform less favorably than TO HAR-STD. Upon closer examination of the computed model confidence sets presented in Table 4, an intriguing pattern emerges. Although the TO HAR-STD and TO FNN-STD models show higher forecast accuracy under the MSE and QLIKE, their counterparts using the extended predictor sets are more frequently included in the model confidence sets of individual sample periods in terms of MSEs and QLIKE losses, even in the initial sample period. This indicates that the primary reason for the inferior overall performance of the TO HAR-EXT and TO FNN-EXT models is largely due to substantial MSEs (QLIKE losses) for one or a limited number of new issues/spinoffs. Consequently, this scenario can lead to instances where the average MSE (QLIKE) ratios of two forecasting models are greater than one, regardless of which of the two models serves as the reference model. For example, we observe this circumstance for the average MSE ratios of the TO XGB-STD and TO-HAR EXT models or the NP FNN-EXT and TO HAR-EXT models.

In summary, our results highlight that complementary source data and transfer learning, in the form of NP and MTL models, lead to significant improvements in the forecast accuracy of both linear and non-linear realized variance forecasting models for new issues and spinoffs. Moreover, our findings demonstrate that our proposed MTL approach, particularly when combined with the extended predictor set, outperforms models trained on the entire source data set via naive pooling (NP), target-only models (TO), and the random walk forecast, irrespective of the amount of available target data. In addition, most MTL models based on the standard predictor set outperform the NP STD and TO STD models under the MSE.

6.2. Characteristics of source data subsequences selected by MTL models

In order to gain a deeper understanding of the source subsequences selected by our proposed transfer learning approach, we conduct an additional analysis on the training data sets of the MTL-25, MTL-50, and MTL-75 models. This analysis examines the distribution of selected subsequences across the complementary source assets and their temporal distance from the forecast origin.

6.2.1. Distribution of selected source subsequences across complementary assets

The analysis of the origins of selected source subsequences, specifically the source assets from which these series were selected, is presented in Tables F.12–F.14 in Appendix F. The tables report, for each target asset, the average selection rates of available subsequences from individual source assets with the average taken across all estimation steps and sample periods. Since the total number of selected subsequences varies depending on the choice of ϵ, the results are presented accordingly. Specifically, Table F.12 reports the respective results for the MTL-25 models, Table F.13 for MTL-50, and Table F.14 for MTL-75.

A closer examination of the selection rates reveals that certain sectors and specific source assets exhibit lower proportions of subsequences included in the training data sets of the MTL models, while others show higher selection rates. For instance, we observe that subsequences of the Consumer Staples sector are percentage-wise less frequently selected for the training sets of certain MTL models, particularly the MTL-25, MTL-50, and MTL-75 models for the target assets TWTR and MRNA, and the MTL-50 models for NCLH, PSX, SYF, and CTVA. Consequently, this implies that the lagged volatility component subsequences of Consumer Staples sector assets exhibit a notable dissimilarity from the volatility component series of the aforementioned target assets. In addition, smaller proportions of subsequences from the majority of Utilities sector assets are selected for training data sets of TWTR and all MRNA MTL models.

Conversely, the training data sets for the TWTR MTL-25 and MTL-50 models exhibit an above-average inclusion rate of subsequences from assets in the Communication Services, Consumer Discretionary, and Information Technology sectors. This pattern of higher inclusion rates extends to the TWTR MTL-75 training data set, with respect to Energy sector assets. Similarly, the MRNA MTL-25, MTL-50, and MTL-75 training sets show a high selection rate of subsequences from the Communication Services, Consumer Discretionary, and Information Technology sector assets. Furthermore, the MRNA target data tends to display a higher realized variance structure similarity with specific Energy sector assets, as indicated by larger proportions of selected subsequences from Energy sector assets in the MRNA MTL-25, MTL-50, and MTL-75 training data sets. A comparable trend is also evident in the PSX MTL-25 and MTL-75 training sets.

A review of the inclusion rates within the respective GICS sectors of each target asset reveals generally moderate selection rates. Notable exceptions are the high inclusion rates of the TWTR MTL-25, TWTR MTL-50, PSX MTL-25, and INVH MTL-50 training data sets relative to their respective sectors.

Aside from the observations discussed, we note that certain source assets demonstrate lower selection rates across a large number of target MTL model training sets. This is evident in cases such as PG for all MTL models; JNJ for the MTL-25 and MTL-50 models; TTWO, TSLA, and DXCM for the MTL-50 and MTL-75 models; and FTNT and QRVO for the MTL-75 models.

Interestingly, our analysis suggests that selecting source subsequences based on the similarity to recent target observations at the forecast origin does not necessarily result in high selection rates for assets belonging to the same sector as the target asset. Although the corresponding sector seems to be relevant, the assets from which subsequences are selected are spread across all sectors considered. However, for specific target assets, certain source assets and sometimes even entire source asset

sectors appear to exhibit dissimilar patterns in their lagged volatility predictors.

6.2.2. Temporal distance between source subsequence extraction points and forecast origins

In addition to analyzing the selection rates of individual complementary assets, we examine the temporal discrepancy between the forecast origins and the points in time from which the selected subsequences are extracted. Specifically, for a given target asset and a given forecast origin, we analyze how many selected subsequences have a start date within the first 100 observations prior to the forecast origin, how many start dates fall within the interval of 101 to 200 observations prior to the forecast origin, and so forth. We set each of these counts in relation to the number of source subsequences available in the respective 100-day interval, i.e. all the source subsequences with a start date that falls in that interval. We repeat this for each forecast origin and compute the average of the corresponding ratios for each 100-day interval. The results are illustrated in Figs. G.4–G.6 of Appendix G. It is important to note that the results for the group of subsequences with the greatest temporal distance from the forecast origin are primarily driven by the ratio of selected subsequences in the final re-estimation steps. Consequently, the average is computed from fewer observations, making it more sensitive to variations in the relative quantity of selected subsequences. Therefore, the significance of this class, i.e. the final bars of each asset chart in Figs. G.4–G.6, should be interpreted with caution, and we refrain from including these groups in our subsequent discussion.

When assessing the selection rates across different temporal distances between the forecast origins and selected subsequences, distinct patterns emerge for different target assets. For example, the MTL models for TWTR and MRNA show a tendency to select subsequences from more distant points in the past relative to the forecast origin, although MRNA also demonstrates high inclusion rates for subsequences in temporal proximity to the forecast origin. In contrast, MTL models for LW and INVH display a preference primarily for subsequences closer to the forecast origin. Meanwhile, MTL models for NCLH, PSX, SYF, CARR, DXC, and CTVA exhibit a relatively stable selection rate for subsequences, irrespective of their temporal proximity to the forecast moment. These findings suggest that target observations close to the forecast origin show high similarities not only to source subsequences from the same time period but also to those that are significantly further back in time. Consequently, this indicates that these far-back source subsequences also enhance the realized variance forecasts for new issues and spinoffs.

6.3. Comparison to HAR-based extensions

The results reported in Section 6.1 demonstrate that our proposed multisource transfer learning (MTL) approach outperforms models trained on the entire target and source data set (naive pooling), target-only (TO) models, and random walk (RW) forecasts. This improvement in forecast accuracy arises from the data-driven selection of relevant source subsequences and their integration with the corresponding target data. Among the evaluated volatility models, the XGBoost implementation with transfer learning and an extended predictor set (MTL XGB-EXT) yields the strongest predictive performance. However, the computational costs of these forecasts[9] motivate an evaluation of whether less computationally demanding models can achieve comparable accuracy, potentially challenging the need for such added complexity.

To investigate this, we further assess whether relatively simple extensions of the HAR framework can deliver competitive performance. Specifically, we compare four HAR-based variants: (i) semivariance HAR (HAR-RS), which substitutes the daily realized variance predictor in the standard HAR model with separate daily positive and negative semivariance predictors (Patton & Sheppard, 2015); (ii) HARQ, which accounts for measurement errors in realized volatility (Bollerslev et al., 2016); (iii) HAR-LASSO, which uses LASSO for variable selection from the standard (STD) or extended (EXT) predictor set (Tibshirani, 1996); and (iv) HAR-Mean, which supplements the HAR model with the equally weighted average of the previous day's realized variances across all source assets and the target asset. In addition, we evaluate the TO HAR-STD model re-estimated daily (HAR-1), rather than every five trading days as in Section 6.1, to examine whether more frequent re-estimation improves its forecasting accuracy.

While all these models are estimated exclusively on target data, the HAR-Mean model also incorporates information from the source assets via an additional regressor constructed from pooled target and source data. Conceptually, this specification can be viewed as a factor-augmented regression model, where (latent) factors derived from a panel of observable variables are used to forecast a target variable. Such factor-based approaches are widely applied in macroeconomic and financial time series forecasting (Bernanke et al., 2005). For instance, Kim and Baek (2019) propose a factor-augmented HAR model to predict the realized volatility of global equity markets. Similarly, Zhang et al. (2020) employ a global diffusion index, calculated as the simple average of realized volatilities across 22 international stock markets, to forecast realized volatility in each market. They also consider a

[9] As an example, consider generating forecasts for CARR, in our setup, 50 days after its initial trading day. Computing the DTW distances requires approximately 1.5 s to compare roughly 7000 source subsequences (each containing 22 observations) against a single 22-observation CARR target sequence. Incorporating these selected source subsequences into the training data further increases the computational load during model estimation. For the MTL-75 specification, this process adds approximately 115,500 training samples (75% of the 7000 subsequences). Using 29 CARR target observations together with the selected subsequences, the corresponding model estimation times range from 0.03 s for MTL-75 HAR-STD to 50.04 s for MTL-75 FNN-EXT (trained over 500 epochs). Notably, XGBoost models are more computationally efficient, requiring at most 1.67 s for MTL-75 XGB-EXT. All computations were executed on CPUs using a four-core Intel Core i7 processor. Although the increased computational load compounds under frequent re-estimation, hyperparameter tuning, or when incorporating additional source data, the measured computation times demonstrate that instance-based transfer learning is feasible even on consumer-grade hardware.

Table 5
Performance of HAR-based extensions and the daily re-estimated TO HAR-STD model compared with the TO HAR-STD model (re-estimated every five days) and the best-performing MTL model, i.e. MTL-75 XGB-EXT. Reported are the cross-sectional average MSEs (QLIKE losses) from rolling one-day-ahead forecasts for each HAR-based extension and the daily re-estimated TO HAR-STD model relative to both the TO HAR-STD (five-day re-estimation) and the MTL-75 XGB-EXT benchmark. The MSEs (QLIKE losses) are based on the entire evaluation period ($s = s^*$), with the MSE (QLIKE loss) criteria of the best-performing model for each benchmark marked in bold. '> 99' represents values that exceed 99. Model names reflect both the forecasting approach and predictor set: 'TO' denotes target-only models trained solely on the target data; the second part of the name indicates the base model class, with 'HAR' referring to the heterogeneous autoregressive model, suffix '1' indicates that the HAR model has been re-estimated daily, 'LASSO' refers to the least absolute shrinkage and selection operator, 'HAR-RS' denotes the semivariance HAR model, and 'HARQ' the quarticity-based extension of the HAR model. HAR-Mean extends the HAR model by the equally weighted average of the previous day's realized variances across all source assets and the target.

Model	Relative to TO HAR-STD		Relative to MTL-75 XGB-EXT	
	MSE	QLIKE	MSE	QLIKE
TO HAR-1-STD	**0.995**	**0.987**	**1.228**	**1.247**
TO HAR-1-EXT	> 99	4.262	> 99	3.778
TO HAR-LASSO-STD	1.000	1.000	1.239	1.265
TO HAR-LASSO-EXT	> 99	1.954	> 99	2.343
TO HAR-RS-STD	1.021	1.009	1.266	1.280
TO HAR-RS-EXT	> 99	1.832	> 99	2.352
TO HARQ-STD	1.010	1.115	1.258	1.452
TO HARQ-EXT	> 99	1.797	> 99	2.348
HAR-Mean-STD	1.008	1.382	1.239	1.879
HAR-Mean-EXT	> 99	1.571	> 99	1.956

HAR-PC specification, where the factor is defined as the first principal component of the realized volatility panel.

Because this section focuses on evaluating the performance of computationally less intensive models, we adopt the simplest factor-augmented HAR specification.[10] Thus, we extend the standard HAR model by including the equally weighted average of the previous day's realized volatilities of the source and target assets. Importantly, this HAR-Mean specification differs from naive pooling and our transfer learning approach, as the source data are incorporated only through a single factor/predictor, rather than by estimating models on pooled target and source data or selected source subsequences.

To evaluate the performance of these HAR extensions, we benchmark them against the TO HAR-STD and against the best-performing model from Section 6.1, the MTL-75 XGB-EXT. The comparison is conducted over the entire evaluation period ($s = s^*$). The results, presented in Table 5, reveal that only the TO HAR-STD re-estimated daily outperforms the five-day re-estimation variant in terms of both the MSE and QLIKE. All other STD HAR extension models deliver comparable or slightly inferior performance under the MSE and perform significantly inferior under QLIKE. Consistent with our earlier findings, TO models using the extended predictor set exhibit notably weaker performance. The weak performance of HAR-based extensions for new issues/spinoffs is most plausibly explained by the limited amount of target data available. While extensions such as HAR-RS, HARQ, HAR-LASSO, and HAR-Mean are beneficial in longer samples, their additional predictors can lead to overfitting. This effect is particularly pronounced for models using the extended predictor set.

The results also indicate that neither the standard predictor set nor the extended predictor set HAR extensions surpass the performance of the MTL-75 XGB-EXT. This provides evidence that the additional computational demands of combining multisource transfer learning with machine learning models are justified, given the substantial gains in predictive accuracy.

6.4. Forecast evaluation in the immediate vicinity of the first trading day

In the forecast assessment presented in the previous subsections, we initiated the forecast analysis 50 days after the first trading day of an asset. However, recognizing that investors may have an interest in volatility forecasts from the very beginning of an asset's life cycle, we conduct a second forecast evaluation to assess the capabilities of our multisource transfer learning approach when faced with extreme target data scarcity. In particular, we evaluate the performance of the introduced realized variance forecasting models for new issues and spinoffs, starting immediately after their first trading day, with the first out-of-sample forecast corresponding to the realized variance of the third trading day. To this end, we report the MSE (QLIKE) of each model relative to the MSE (QLIKE) of the TO HAR-STD as a cross-sectional average

[10] According to Zhang et al. (2020), both HAR-Mean and HAR-PC outperform alternative model specifications, but the mean diffusion index tends to be more efficient, supporting our decision to focus on HAR-Mean.

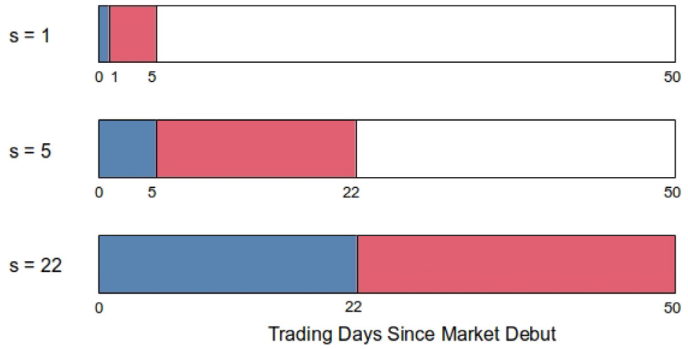


Fig. 3. Training and evaluation data sets for new issues and spinoffs immediately following their first trading day across different sample periods ($s = 1$, $s = 5$, and $s = 22$). Blue intervals represent exclusive training periods, while red intervals indicate evaluation periods during which all realized variance forecasting models are re-estimated daily in this setting.

starting from the first day after the distribution date up to 50 days after the initial trading day. We segment this time period into three sample periods, as illustrated in Fig. 3. The first sample period ($s = 1$) covers the initial four trading days following the first trading day. The second sample period ($s = 5$) extends from the sixth to the 22nd trading day, while the third sample period ($s = 22$) spans from the 23rd to the 50th trading day post-distribution. The evaluation set sizes for the specified periods are four, 17, and 28 days, respectively. The forecasts are evaluated on a daily basis using a rolling approach, where each forecast is generated for the subsequent trading day.

Consequently, for the forecasts of the first two sample periods, i.e. after one and five days, we must exclude predictors that represent lagged aggregates or differenced values such as the weekly and monthly volatility components or the *MOM* and *DV* predictors. We introduce a corresponding notation for the model specification: the forecasting models of Section 6.1, that use the entire set of predictors in Q_{std} and Q_{ext}, are denoted by Q22. Specifications that exclude the monthly volatility component and *MOM* predictor are denoted by Q5, while those lacking the RV_m, RV_w, *MOM*, *DV*, *US3M*, and *HSI* predictors are denoted by Q1. The application of the respective models to the sample periods commences as soon as a sufficient number of target observations are available. Specifically, the Q1 model specifications are utilized immediately following the first trading day and are applied to all subsequent sample periods ($s = 1$, $s = 5$, and $s = 22$). Similarly, the Q5 model specifications are initiated after five trading days, and then applied consistently throughout the following sample periods ($s = 5$ and $s = 22$), while specifications with the complete predictor set, i.e. the Q22, are applied only to the last sample period, i.e. $s = 22$, starting after 22 trading days.

In this forecasting setup, the models are re-estimated daily to address the extremely limited quantity of target observations. The length of source subsequences for MTL models is dynamically matched to the length of the available target data, up to a maximum of 22 days, at each re-estimation point, while the DTW distance between the target and source subsequences is determined based only on the available lagged volatility predictors of the respective predictor sets. In addition, the TO FNNs are trained with a batch size of one and, if only one target observation is available, with a fixed epoch count of 500.

We acknowledge that the inclusion of TO models may be highly controversial due to their limited training data. Nevertheless, we include TO models in our analysis for the purpose of comparison and to maintain consistency.

6.4.1. Realized variance forecasting up to 50 days after the first trading day

Table 6 reports the MSEs (QLIKE losses) of the forecasting models relative to the MSE (QLIKE) of the TO HAR-STD model using the Q1 specification. The results indicate that the benchmark TO HAR-STD Q1 specification underperforms the RW at $s = 1$, performs comparably at $s = 5$, and outperforms it at $s = 22$. Most other TO models, apart from the TO XGB Q1 specification, also exhibit substantially higher MSEs and QLIKE losses than the RW, particularly in their initial sample periods. This is not entirely surprising, considering that the TO models with Q1 specification are estimated on a maximum of just 49 data points at the last forecast origin, and those with Q5 and Q22 specifications are trained on only up to 45 and 28 observations, respectively.

Our results further show that most models incorporating additional source data in their training sets, namely the NP and MTL specifications, outperform the RW on average across all sample periods. Similarly, these models generally exceed the performance of the TO HAR-STD model with the Q1 specification in the first sample period in which they become available. This pattern holds for the Q1 models, which use the most restricted predictor set, as well as for the Q5 models introduced after five trading days. For the Q22 models, however, forecast improvements are less consistent: while certain NP and MTL models still outperform the TO HAR-STD model with

Table 6
One-day-ahead cross-sectional average relative MSEs and QLIKE losses by model, sample period $s \in \{1, 5, 22\}$, and specification (Q1, Q5, Q22). Reported are the MSEs (QLIKE losses) of forecasting models relative to the MSE (QLIKE) of the TO-HAR-STD model with Q1 specification, averaged across all new issues and spinoffs. The model specification (Q1, Q5, Q22) indicates which predictors from the STD and EXT sets (as denoted by the model name extensions in the first column) are excluded: Q1 excludes RV_m, RV_w, *MOM*, *DV*, *US*3*M*, and *HSI*; Q5 excludes RV_m and *MOM*, and Q22 excludes none. The MSE and QLIKE error metrics are computed based on rolling one-day-ahead forecasts over the three sample periods (s) illustrated in Fig. 3. '—' means not applicable. The best-performing model for each sample period (s) and error metric within each specification (Q1, Q5, Q22) is marked in blue, while the best-performing model across all specifications for each sample period and error metric is highlighted in bold blue. '> 99' represents values that exceed 99.

	$s=1$						$s=5$						$s=22$					
	MSE			QLIKE			MSE			QLIKE			MSE			QLIKE		
Model \ Spec.	Q1	Q5	Q22	Q1	Q5	Q22	Q1	Q5	Q22	Q1	Q5	Q22	Q1	Q5	Q22	Q1	Q5	Q22
RW	0.736	–	–	0.908	–	–	1.050	–	–	1.017	–	–	1.273	–	–	1.443	–	–
TO HAR-STD	1.000	–	–	–	–	–	1.000	1.900	–	1.000	1.844	–	1.000	1.088	17.529	1.000	1.183	2.438
TO HAR-EXT	2.225	–	–	0.974	–	–	> 99	10.601	–	4.136	2.614	–	3.305	2.465	58.925	2.773	6.713	10.596
NP HAR-STD	0.760	–	–	0.704	–	–	0.964	0.861	–	0.861	0.702	–	1.214	0.981	0.963	1.240	1.013	0.992
NP HAR-EXT	0.672	–	–	0.756	–	–	0.992	0.916	–	0.937	0.775	–	1.566	1.380	1.408	1.275	1.077	1.062
MTL-25 HAR-STD	0.564	–	–	**0.632**	–	–	0.904	0.816	–	1.038	0.812	–	1.013	0.934	0.902	1.461	1.134	1.031
MTL-50 HAR-STD	**0.549**	–	–	0.667	–	–	0.924	0.816	–	1.139	0.793	–	1.053	0.939	0.895	1.668	1.156	1.012
MTL-75 HAR-STD	**0.549**	–	–	0.699	–	–	0.935	0.813	–	1.185	0.774	–	1.077	0.940	**0.891**	1.782	1.151	0.996
MTL-25 HAR-EXT	0.616	–	–	0.754	–	–	0.887	0.852	–	1.069	0.859	–	1.091	1.001	0.993	1.654	1.195	1.037
MTL-50 HAR-EXT	0.612	–	–	0.812	–	–	0.902	0.855	–	1.186	0.850	–	1.157	1.008	0.960	1.970	1.235	1.004
MTL-75 HAR-EXT	0.616	–	–	0.854	–	–	0.917	0.852	–	1.264	0.839	–	1.190	1.005	0.941	2.156	1.232	0.986
TO FNN-STD	4.884	–	–	> 99	–	–	1.579	2.009	–	> 99	> 99	–	1.271	1.065	2.045	> 99	1.062	> 99
TO FNN-EXT	2.996	–	–	> 99	–	–	1.558	> 99	–	16.201	62.508	–	3.605	2.869	2.523	1.699	1.975	> 99
NP FNN-STD	0.813	–	–	0.712	–	–	0.963	0.851	–	0.863	**0.698**	–	1.220	0.989	0.932	1.242	1.018	**0.960**
NP FNN-EXT	0.657	–	–	0.796	–	–	1.016	0.939	–	0.962	0.796	–	1.648	1.472	1.742	1.319	1.153	1.393
MTL-25 FNN-STD	0.564	–	–	**0.632**	–	–	0.904	0.815	–	1.037	0.813	–	1.013	0.931	0.899	1.461	1.130	1.022
MTL-50 FNN-STD	0.551	–	–	0.668	–	–	0.924	0.819	–	1.137	0.796	–	1.053	0.947	0.895	1.668	1.161	1.011
MTL-75 FNN-STD	0.550	–	–	0.700	–	–	0.936	0.815	–	1.185	0.779	–	1.077	0.940	**0.891**	1.781	1.151	0.997
MTL-25 FNN-EXT	0.619	–	–	0.747	–	–	0.912	0.850	–	1.127	0.860	–	1.066	1.020	1.037	1.544	1.166	1.033
MTL-50 FNN-EXT	0.614	–	–	0.782	–	–	0.923	0.844	–	1.220	0.847	–	1.130	1.015	1.009	1.822	1.212	1.021
MTL-75 FNN-EXT	0.594	–	–	0.732	–	–	0.945	0.840	–	1.267	0.833	–	1.132	1.001	0.977	1.888	1.212	1.007
TO XGB-STD	0.613	–	–	0.792	–	–	0.986	1.028	–	1.035	1.626	–	1.777	1.432	1.307	1.676	1.914	1.664
TO XGB-EXT	0.585	–	–	0.725	–	–	**0.768**	0.900	–	0.700	1.066	–	1.463	1.383	1.193	1.830	1.533	1.439
NP XGB-STD	0.640	–	–	0.708	–	–	0.995	0.851	–	1.149	0.874	–	1.078	0.994	0.994	1.461	1.220	1.244
NP XGB-EXT	0.630	–	–	0.727	–	–	0.947	0.889	–	1.134	1.003	–	1.093	1.032	1.065	1.544	1.258	1.256
MTL-25 XGB-STD	0.619	–	–	0.707	–	–	0.937	0.845	–	1.067	0.830	–	0.985	0.949	0.918	1.358	1.114	1.015
MTL-50 XGB-STD	0.656	–	–	0.751	–	–	0.951	0.864	–	1.130	0.828	–	1.001	0.934	0.904	1.467	1.109	0.975
MTL-75 XGB-STD	0.595	–	–	0.711	–	–	0.961	0.840	–	1.167	0.803	–	0.999	0.924	0.918	1.461	1.092	0.995
MTL-25 XGB-EXT	0.839	–	–	0.803	–	–	0.872	0.843	–	0.994	0.855	–	1.087	1.125	1.046	1.549	1.117	1.027
MTL-50 XGB-EXT	0.802	–	–	0.941	–	–	0.893	0.881	–	1.092	0.858	–	1.096	0.937	1.114	1.634	1.094	1.011
MTL-75 XGB-EXT	0.650	–	–	0.823	–	–	0.894	0.864	–	1.108	0.840	–	1.190	0.978	1.166	1.713	1.142	1.011

the Q1 specification, others achieve only comparable or inferior accuracy at $s = 22$, particularly under QLIKE. Moreover, the relative performance of the Q1-NP, Q1-MTL, Q5-NP, and Q5-MTL models declines in later sample periods compared to the TO HAR-STD model with the Q1 specification. Finally, while nearly all MTL models outperform their NP counterparts under the MSE, this advantage is less pronounced under QLIKE.

Interestingly, in instances where the forecasting periods of the Q1, Q5, and Q22 specifications intersect, there is a noticeable trend for most NP and MTL models to show modest decreases in the relative MSE and QLIKE, favoring the more complex respective model. This observation suggests a viable approach for NP and MTL models to transition to their Q5 and Q22 counterparts (Q5-NP, Q5-MTL, Q22-NP, and Q22-MTL) as soon as a sufficient amount of data become available, specifically after five and 22 trading days, respectively. In the following subsection, to investigate whether adopting more complex models, which are trained on fewer target observations, is advantageous over simpler models that utilize a larger data set of target observations, we examine the effectiveness of different strategies for transitioning from one predictor set to another.

6.4.2. Predictor set transition strategies

In total, we assess three different strategies to transition between the predictor sets of all NP and MTL forecasting models after five and 22 days from the distribution date of the respective target assets. The first strategy, herein referred to as 'Str. 1-1-1', involves uniformly using the predictor sets of the Q1 models that lack RV_m, RV_w, *MOM*, *DV*, *US*3*M*, and *HSI* predictors across all three sample periods. The second approach, denoted as 'Str. 1-5-5', entails transitioning from the predictors of Q1 models to those of Q5 models for the second and third sample periods. The third strategy, 'Str. 1-5-22', advocates for a prompt transition to the more complex predictor sets as soon as adequate data become available, specifically switching to Q5 model predictors after five days and to Q22 model predictors after 22 days. This probably represents the most intuitive approach to exploit the increasing availability of data over time, allowing more predictors to be integrated progressively.

Table 7
Performance of transition strategies. Reported are the averages of the MSEs (QLIKE losses) of the column strategy relative to the MSEs (QLIKE losses) of the row strategy. MSEs (QLIKE losses) are based on 49 one-day-ahead realized variance forecasts following the first trading day of a new issue/spinoff. The averages are computed across all new issues and spinoffs, as well as across all NP and MTL models.

Strategy	Relative MSE			Relative QLIKE		
	Str. 1-1-1	Str. 1-5-5	Str. 1-5-22	Str. 1-1-1	Str. 1-5-5	Str. 1-5-22
Str. 1-1-1	1.000	0.929	0.917	1.000	0.822	0.795
Str. 1-5-5	1.087	1.000	0.987	1.283	1.000	0.956
Str. 1-5-22	1.104	1.015	1.000	1.373	1.053	1.000

Table 8
Performance of transition strategies for TO models compared to the best-performing MTL and NP model strategy (Str. 1-5-22). Reported are the cross-sectional average MSEs (QLIKE losses) for each forecasting model (HAR, FNN, XGB) under different transition strategies, predictor sets (STD, EXT), and forecasting approaches (TO, NP, MTL) relative to the Str. 1-1-1 TO HAR-STD. The MSEs (QLIKE losses) are based on 49 one-day-ahead realized variance forecasts following the first trading day of a new issue/spinoff. '—' means not applicable. The MSE (QLIKE loss) criteria of the best-performing model within each model class are marked in bold. '> 99' represents values that exceed 99.

	MSE relative to Str. 1-1-1 TO HAR-STD			QLIKE relative to Str. 1-1-1 TO HAR-STD		
	HAR	FNN	XGB	HAR	FNN	XGB
Str. 1-1-1 TO-STD	–	1.748	1.039	–	> 99	1.139
Str. 1-1-1 TO-EXT	21.087	1.896	0.899	3.115	> 99	1.022
Str. 1-5-5 TO-STD	1.371	1.908	0.989	1.452	> 99	1.530
Str. 1-5-5 TO-EXT	5.560	> 99	0.940	4.002	> 99	1.070
Str. 1-5-22 TO-STD	12.190	2.224	0.943	2.042	> 99	1.424
Str. 1-5-22 TO-EXT	41.706	> 99	0.869	4.879	> 99	1.050
Str. 1-5-22 NP-STD	0.766	0.750	0.765	**0.746**	**0.730**	0.904
Str. 1-5-22 NP-EXT	0.848	0.914	0.777	0.795	0.904	0.946
Str. 1-5-22 MTL-25-STD	0.719	0.717	0.748	0.807	0.804	0.814
Str. 1-5-22 MTL-25-EXT	0.741	0.746	0.859	0.821	0.820	0.832
Str. 1-5-22 MTL-50-STD	0.716	0.716	0.759	0.796	0.798	0.797
Str. 1-5-22 MTL-50-EXT	0.738	0.741	0.843	0.812	0.820	0.835
Str. 1-5-22 MTL-75-STD	**0.712**	**0.712**	**0.734**	0.784	0.787	**0.792**
Str. 1-5-22 MTL-75-EXT	0.735	0.732	0.785	0.805	0.809	0.818

To analyze the forecasting performance of these transition strategies, we employ a pairwise comparison approach similar to that in Section 6.4.1. Specifically, for each new issue and spinoff, we compute the MSE (QLIKE) of one strategy relative to another for both the MTL and NP models. Table 7 reports the average of these MSE (QLIKE) ratios, where, for clarity, the averages are computed across the cross-section as well as across all MTL and NP models. The findings reveal a distinct performance hierarchy, with Str. 1-5-22 outperforming Str. 1-5-5, which in turn exceeds the performance of Str. 1-1-1. While the improvements in relative MSE and QLIKE for Str. 1-5-22 over Str. 1-5-5 are modest, they nonetheless indicate a benefit from promptly adopting more complex predictor sets. Notably, this pattern is also observed individually when separating the results for the MTL and NP models.[11]

Additionally, we apply all transition strategies to the TO models and compare their forecasting performance against the NP and MTL models using the best-performing NP/MTL transition strategy, i.e. Str. 1-5-22. Table 8 reports for each model the cross-sectional average of the MSE (QLIKE) relative to the Str. 1-1-1 TO HAR-STD benchmark. Our findings demonstrate that, in terms of the MSE, the MTL STD models consistently deliver the most accurate forecasts within each model class, i.e. HAR, FNN, and XGB, with the MTL-75 STD specification exhibiting the strongest predictive performance. With respect to QLIKE, the Str. 1-5-22 MTL-75 XGB STD achieves the highest predictive accuracy among the XGB models, while the Str. 1-5-22 NP STD models yield the highest accuracy for HAR and FNN. Notably, in contrast to the forecast assessment in Section 6.1, which begins 50 days after the first trading day of each new issue/spinoff, the STD predictor set MTL models outperform their EXT predictor set counterparts in terms of the MSE and QLIKE.

To summarize, the examination of realized variance forecasts for new issues and spinoffs commencing on the first day after the distribution date underscores another significant finding: the integration of complementary source data and transfer learning enhances realized variance forecasts even immediately following the distribution date. Furthermore, our results suggest that transitioning to more complex models once a sufficient amount of data are available represents a viable option.

[11] The corresponding results are available from the authors upon request.

7. Conclusion

In this paper, we considered the problem of forecasting the volatility of assets with limited data availability, such as new issues/spinoffs. We argued that complementary data from financial source assets with a comprehensive historical record can improve volatility forecasts for new issues and spinoffs. To this end, we proposed a model-agnostic transfer learning approach based on the DTW distance between recent target observations and selected source data subsequences. In order to compare our multisource instance selection approach with methods that rely solely on either target asset data or target data combined with the entirety of the source data, we conducted a realized variance forecast assessment of 10 new issues and spinoffs. This forecast evaluation incorporated both linear and non-linear models—i.e. HAR models, XGBoost models, and FNNs—and considered HAR-based extensions as additional benchmarks. Moreover, we assessed all forecasting models using the standard HAR predictor set and an extended predictor set. Furthermore, we analyzed the characteristics of source subsequences selected by our transfer learning approach and presented a second forecast assessment in which we evaluated the forecasting precision of the proposed multisource transfer learning approach in the immediate time period after the first trading day. In this way, we extend existing research by considering volatility forecasting for new issues and spinoffs, and the application of transfer learning methods to forecasting realized variance.

Our results highlighted that knowledge transfer from complementary data increased the predictive accuracy of all models considered. In particular, our proposed transfer learning method in combination with XGBoost models showed superior forecasting performance compared to any other model class or training data set approach, including the considered HAR-based extensions. Moreover, we observed higher forecast accuracy for our instance selection approach in combination with the HAR and FNN models compared to models trained on the target data alone or trained using a naive pooling approach. Hence, multisource transfer learning improves the forecast performance irrespective of the adopted model class. The superior performance of our transfer learning approach compared to naive pooling also indicates that the negative information transfer of global modeling approaches can be mitigated by transferring knowledge only from similar instances. Notably, the improvements in HAR models achieved through transfer learning are relevant in practice, as they enable more accurate forecasts without compromising the interpretability that linear models provide. In contrast, neural networks offer little to no interpretability, and XGBoost provides only partial insights, such as feature-importance scores, rather than fully transparent relationships.

The analysis of the subsequences selected by our transfer learning approach revealed that the selected subsequences were limited neither to the GICS sector of the respective target asset nor to more recent source observations. Furthermore, we observed strong variability in the number of selected subsequences for different target assets. This suggests that it is useful to consider a wide range of different source assets with extensive historical data in order to improve the predictive accuracy of new issue/spinoff realized variance forecasting models.

Furthermore, our empirical analysis showed that multisource transfer learning improves the forecast accuracy even in the extreme yet practically important scenario in which volatility forecasts are required immediately after the first trading day of the new issue/spinoff. Our findings confirmed the advantages of integrating source data and transfer learning methods into realized variance forecasting models. The majority of models that leveraged source data provided superior forecasts compared to those trained solely on limited target data or a random walk forecast. In addition, our results suggested that transitioning to more complex models—specifically those incorporating weekly and monthly volatility components—is advantageous as more data become available. The benefits of these models manifest immediately upon their availability.

To the best of our knowledge, this is the first paper to consider forecasting the realized volatility of new issues and spinoffs. Notably, these stocks often exhibit unusually high volatility during their initial trading days or weeks; see e.g. Lowery et al. (2010). These elevated early volatilities commonly stem from intense price discovery, speculative trading, and illiquidity. In addition, newly listed stocks often exhibit lower trading volumes and wider bid–ask spreads in the early phase, which can introduce more pronounced microstructure noise into realized volatility measures and reduce their reliability; see e.g. Aït-Sahalia and Yu (2009) regarding the impact of illiquidity on realized volatility measures. Volatility is also shaped by investor sentiment, as much as by fundamental economic factors (Vamossy, 2025). Moreover, because early stage listings lack a stable investor base or trading rhythm, their reactions to news tend to be more unpredictable. All of these dynamics may require the development of specific volatility models whose estimation and application, however, is hampered by the limited data availability. As shown in this paper, transfer learning can provide a solution to this problem. In particular, combining transfer learning with popular realized volatility models yields substantial improvements in forecasting accuracy for new issues/spinoffs. In future research, it will be interesting to develop volatility models that are tailored to capturing the particularities of these assets and to combine them with transfer learning applied exclusively among new issues/spinoffs. Naturally, this would come at the cost of relying on a much smaller source data set.

Further research in the area of incorporating complementary data and transfer learning is also warranted to explore the full potential of these methods to enhance the precision of volatility forecasting models for new issues and spinoffs. This may involve an extensive hyperparameter optimization or the application of different forecasting models such as transformer networks. Moreover, our results motivate an assessment of how complementary source data can be used to improve volatility predictions for target assets that already have an extensive data history, i.e. multiple years of available data.

Data and code availability

The high-frequency stock market data used in this study was sourced from FirstRate Data LLC (https://www.firstratedata.com). Access to this data is subject to licensing restrictions; users must purchase the data directly from the provider, as redistribution is not permitted. Earnings announcement dates were obtained from Yahoo Finance (https://finance.yahoo.com). Macroeconomic and financial uncertainty indicators were collected from publicly available sources, including the U.S. 3-month Treasury bill rate (US3M) from the Federal Reserve Bank of St. Louis FRED database (https://fred.stlouisfed.org/series/DTB3); the Aruoba-Diebold-Scotti business conditions index (ADS) from the Federal Reserve Bank of Philadelphia (https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/ads); the economic policy uncertainty (EPU) index (https://www.policyuncertainty.com); the CBOE Volatility Index (VIX) historical data (https://www.cboe.com); and the Hang Seng index (HSI) obtained via Yahoo Finance.

The source code is publicly available at https://github.com/a-teller/vola-mtl.

CRediT authorship contribution statement

Andreas Teller: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Uta Pigorsch:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Christian Pigorsch:** Writing – review & editing.

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Appendix A. Dynamic time warping

Consider a time series $T = (\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_n, \ldots, \mathbf{t}_N)$ with $N \in \mathbb{N}$ and $\mathbf{t}_n \in \mathbb{R}^p$, and a time series $S = (\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_m, \ldots, \mathbf{s}_M)$ with $M \in \mathbb{N}$ and $\mathbf{s}_m \in \mathbb{R}^p$. In order to align both series, DTW constructs a local cost matrix $C \in \mathbb{R}^{MxN}$, defined as

$$C(n, m) := c(\mathbf{t}_n, \mathbf{s}_m), \tag{A.1}$$

where each cell represents the cost (c)—i.e. the distance, typically the Euclidean distance, between two observations of T and S. The optimal alignment between the two time series is identified by finding the minimum-cost warping path through the cost matrix. We define a warping path O as a sequence of points $O = (o_1, o_2, \ldots, o_L)$ with $o_l = (o_i, o_j)$, $i \in N, j \in M$, and $l \in L$, which adheres to the following constraints:

- Boundary condition: $o_1 = (1, 1)$ and $o_L = (N, M)$
- Monotonicity condition: $n_1 \leq n_2 \leq \cdots \leq n_L$ and $m_1 \leq m_2 \leq \cdots \leq m_L$
- Continuity condition: $o_{l+1} - o_l \in \{(1, 1), (1, 0), (0, 1)\}$

While the boundary condition ensures that the warping path starts at the beginning of both sequences and ends at the end of both sequences, the monotonicity condition prevents the warping path from moving backward in time for either sequence. Additionally, the continuity condition defines the step size of the warping path to avoid overly large jumps in one sequence relative to the other. The total cost of a warping path with respect to the cost matrix is defined by

$$c_O = \sum_{l=1}^{L} c(\mathbf{t}_{n_l}, \mathbf{s}_{m_l}), \tag{A.2}$$

where $(\mathbf{t}_{n_l}, \mathbf{s}_{m_l})$ is a matrix element which also represents the lth element of the warping path O. As shown by Bellman and Kalaba (1959), an optimal warping path, which minimizes Eq. (A.2), can be determined recursively by dynamic programming. The DTW distance $DTW(T, S)$ between T and S is then defined as the total cost of the optimal path O^*:

$$DTW(T, S) = c_{O^*} = \sum_{l=1}^{L} c(\mathbf{t}_{n_l}, \mathbf{s}_{m_l}). \tag{A.3}$$

Appendix B. Source data set

See Table B.9.

Appendix C. Feedforward neural networks

FNNs are based on the interconnection of multiple artificial neurons, i.e. computational units, that perform simple calculations in the form of weighted aggregations and (non-linear) transformations. These artificial neurons can be defined by

$$a = f\left(\sum_{j=1}^{J} w_j z_j + b\right), \tag{C.1}$$

where the weights, w_j, and the bias terms, b, are trainable parameters; z_j denotes the inputs of the artificial neuron; J denotes the number of inputs; f is an arbitrary activation function; and a is the activation, i.e. the output of the artificial neuron.

In FNN frameworks, artificial neurons are arranged in multiple layers. The input layer receives training data observations and passes them on to the next layer. The output layer produces the final prediction of the network. Intermediate layers are commonly referred to as hidden layers. In general, for an L-layer FNN, the output of the lth layer is given by

$$\mathbf{a}^l = f^l(\mathbf{W}^l \mathbf{a}^{l-1} + \mathbf{b}^l), \quad 1 \leq l \leq L, \tag{C.2}$$

Table B.9
Overview of source data set assets ordered by GICS sectors.

Ticker	Name	GICS sector
DISH	Dish Network	Communication Services
EA	Electronic Arts Inc.	Communication Services
GOOGL	Alphabet Inc.	Communication Services
LUMN	Lumen Technologies Inc.	Communication Services
META	Meta Platforms Inc.	Communication Services
TTWO	Take Two Interactive Software Inc.	Communication Services
AMZN	Amazon.com Inc.	Consumer Discretionary
EBAY	eBay Inc.	Consumer Discretionary
LEN	Lennar Corp.	Consumer Discretionary
MHK	Mohawk Industries	Consumer Discretionary
NWL	Newell Brands	Consumer Discretionary
TSLA	Tesla Inc.	Consumer Discretionary
CPB	Campbell Soup Company	Consumer Staples
HSY	Hershey Company	Consumer Staples
KMB	Kimberly-Clark Corp.	Consumer Staples
PG	Procter & Gamble Company	Consumer Staples
TAP	Molson Coors Beverage Company	Consumer Staples
WMT	Walmart Inc.	Consumer Staples
APA	Apache Corporation	Energy
CVX	Chevron Corporation	Energy
DVN	Devon Energy Corp. 66	Energy
EQT	EQT Corp.	Energy
HES	Hess Corp.	Energy
XOM	ExxonMobil	Energy
AIZ	Assurant Inc.	Financials
BRK-B	Berkshire Hathaway Inc.	Financials
JPM	JPMorgan Chase & Co.	Financials
LNC	Lincoln National Corp.	Financials
MTB	M&t Bank Corp.	Financials
NDAQ	Nasdaq Inc.	Financials
BIIB	Biogen	Health Care
DVA	DaVita HealthCare Partners Inc.	Health Care
DXCM	DexCom Inc.	Health Care
JNJ	Johnson & Johnson	Health Care
UNH	UnitedHealth Group Inc.	Health Care
XRAY	Dentsply Sirona Inc.	Health Care
ALK	Alaska Air Group	Industrials
FAST	Fastenal Company	Industrials
GNRC	Generac Holdings Inc.	Industrials
GWW	W. W. Grainger	Industrials
HON	Honeywell International Inc.	Industrials
UPS	United Parcel Service of America Inc.	Industrials
AAPL	Apple Inc.	Information Technology
FFIV	F5 Inc.	Information Technology
FTNT	Fortinet	Information Technology
MSFT	Microsoft Corp.	Information Technology
QRVO	Qorvo	Information Technology
TEL	TE Connectivity Ltd.	Information Technology
APD	Air Products & Chemicals Inc.	Materials
IFF	International Flavors & Fragrances Inc.	Materials
LIN	Linde plc	Materials
SEE	Sealed Air Corp.	Materials
VMC	Vulcan Materials Company	Materials
WRK	WestRock Company	Materials
AMT	American Tower Corp.	Real Estate
AVB	AvalonBay Communities Inc.	Real Estate
FRT	Federal Realty Investment Trust	Real Estate
PLD	Prologis Inc.	Real Estate
VNO	Vornado Realty Trust	Real Estate
WY	Weyerhaeuser Company	Real Estate
AWK	American Water Works Company	Utilities
DUK	Duke Energy Corp.	Utilities
ES	Eversource Energy	Utilities
NEE	NextEra Energy Inc.	Utilities
NRG	NRG Energy	Utilities
PNW	Pinnacle West Capital	Utilities

where $\mathbf{a}^l \in \mathbf{R}^{U_l}$ is the activation vector of layer l, consisting of U_l artificial neurons; $\mathbf{a}^{l-1} \in \mathbf{R}^{U_l}$ is the activation vector of the previous layer, $l-1$, consisting of U_{l-1} units; $\mathbf{W}^l \in \mathbf{R}^{U_l \times U_{l-1}}$ is the weight matrix; $\mathbf{b}^l \in \mathbf{R}^{\mathbf{U_l}}$ is the bias vector; and f^l is the activation function of layer l. Due to the interconnection of artificial neurons and their non-linear transformations, FNNs with at least one hidden layer are capable of modeling arbitrarily complex non-linear dependency structures (Gybenko, 1989).

The number of artificial neurons in the input layer is specified by the number of input features. The number of hidden layers and units within them represent tunable hyperparameters. Similarly, the class of activation functions applied in individual hidden layers also constitutes a tunable hyperparameter. The number of output neurons and their activation functions, on the other hand, are determined by the specific task for which the model is designed. Regression tasks, such as realized variance forecasting, typically require a single output neuron with an identity activation function. The training of FNNs is conducted by adjusting the trainable parameters θ, i.e. the connection weights and bias terms, to minimize a cost function C, which can be, for instance, the mean squared error loss function:

$$C(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - f_\theta(\mathbf{x}_i))^2, \tag{C.3}$$

where $\mathbf{x}_i$ denotes the ith training example, y_i is the corresponding true label, and n is the number of training examples. To minimize C, gradient descent or its variants such as Adagrad (Duchi et al., 2011) or ADAM optimization (Kingma & Ba, 2015) are used in combination with the backpropagation algorithm (Rumelhart et al., 1986). Backpropagation calculates the gradient of the loss function by propagating the model error, i.e. the difference between the predicted output and the true labels, from the last layer to the first layer, while using the chain rule of calculus to determine the partial derivatives of each parameter with respect to C. The parameters are then updated by the respective optimization algorithm based on the gradient.

Appendix D. XGBoost

For a given set of observations $\{\mathbf{x}_i, y_i\}$ of size n with $\mathbf{x}_i \in \mathbf{R}^p$ and $y_i \in \mathbf{R}$, consider a single regression tree that predicts according to

$$f(\mathbf{x}_i) = \sum_{k=1}^{K} w_k I(\mathbf{x}_i \in R_k), \tag{D.1}$$

where K is the number of terminal nodes (leaves), w_k is the leaf weight of the kth leaf, and I is an indicator function that is one if $\mathbf{x}_i$ falls into the region R_k and zero otherwise. The XGBoost model represents an ensemble of such trees, and the predictions are computed according to

$$F(\mathbf{x}_i) = \sum_{m=1}^{M} \epsilon f_m(\mathbf{x}_i), \quad f_m \in F, \tag{D.2}$$

where f_m is the mth regression tree, M is the total number of boosting iterations (trees), ϵ is a shrinkage parameter, and F is the space of regression trees.

The estimation of this composite model is conducted in a stage-wise manner. In each boosting iteration m, a new regression tree f_m is added to the existing ensemble and constructed in such a way that it minimizes the following objective:

$$Obj_m = \sum_{i=1}^{n} l(y_i, F_{m-1}(\mathbf{x}_i) + f_m(\mathbf{x}_i)) + \Omega(f_m), \tag{D.3}$$

where l is a differentiable convex loss function, F_{m-1} is the ensemble model up to the $(m-1)$-th iteration, and $\Omega(f_m)$ is a regularization term. The regularization component, which penalizes the size of the leaf weights (L2 regularization) and the number of terminal nodes, is given by

$$\Omega(f_m) = \gamma K_m + \frac{1}{2}\lambda \sum_{k=1}^{K_m} w_{m,k}^2, \tag{D.4}$$

where γ and λ are regularization parameters, K_m is the number of terminal nodes of the mth tree, and $w_{m,k}$ represents the leaf weights of the tree. In contrast to gradient boosting methods, which estimate weak learners with respect to the negative gradient, XGBoost approximates Obj_m using second-order Taylor expansion, i.e. Newton boosting, with

$$Obj_m \simeq \sum_{i=1}^{n}\left[l(y_i, F_{m-1}(\mathbf{x}_i)) + g_i f_m(\mathbf{x}_i) + \frac{1}{2}h_i f_m^2(\mathbf{x}_i)\right] + \Omega(f_m), \tag{D.5}$$

where g_i and h_i are the first- and second-order derivatives of the loss function with respect to the predictions of F_{m-1}. Empirical evidence suggests that this higher order of approximation combined with regularization leads to more accurate estimates of Eq. (D.2) compared to vanilla gradient tree boosting (Sigrist, 2021).

For each tree f_m, after removing the constant terms and considering the leaf weights, the objective simplifies to

$$\bar{Obj}_m = \sum_{k=1}^{K_m}\left[\left(\sum_{i=1}^{n} g_i I(\mathbf{x}_i \in R_{m,k})\right) w_{m,k} + \frac{1}{2}\left(\sum_{i=1}^{n} h_i I(\mathbf{x}_i \in R_{m,k}) + \lambda\right) w_{m,k}^2\right] + \gamma K_m. \tag{D.6}$$

When differentiating Eq. (D.6) with respect to $w_{m,k}$, the optimal weight $w_{m,k}^*$ for a given structure $q(\mathbf{x})$ is defined by

$$w_{m,k}^* = -\frac{\sum_{i=1}^{n} g_i I(\mathbf{x}_i \in R_{m,k})}{\sum_{i=1}^{n} h_i I(\mathbf{x}_i \in R_{m,k}) + \lambda}. \tag{D.7}$$

Moreover, substituting the leaf weights of Eq. (D.6) by the optimal leaf weights of Eq. (D.7) provides a criterion for evaluating and comparing different tree structures within

Table E.10
One-day-ahead cross-sectional average relative MSEs. Each value in this table represents the cross-sectional average of the pairwise realized variance forecast MSE for the model in the selected column relative to the benchmark in the selected row. Each model's MSE is determined for one-day-ahead realized variance forecasts over 500 days, starting 50 days after the first trading day of each target asset. The symbol (∗) denotes that the Diebold–Mariano test of equal predictive accuracy is rejected for more than 50% of the target assets using a 5% significance level. A rejection for the majority of target assets indicates that the model in the column exhibits a significantly lower MSE than the corresponding benchmark model in the respective row. The corresponding MSE values are additionally marked in blue. '—' means not applicable. Model names reflect both the forecasting approach and predictor set: TO denotes models trained solely on the target data; NP denotes naive pooling, where models are trained on both target and all available source data; and MTL denotes multisource transfer learning, where the numerical suffix (25, 50, 75) indicates the percentage of subsequences transferred from the source data into the final training data set. The second part of the name indicates the base model class, with HAR referring to the heterogeneous autoregressive model, FNN to feedforward neural networks, and XGB to XGBoost, while STD refers to the standard HAR predictor set, and EXT to an extended predictor set.

	RW	TO HAR-STD	NP HAR-STD	MTL-25 HAR-STD	MTL-50 HAR-STD	MTL-75 HAR-STD	TO FNN-STD	NP FNN-STD	MTL-25 FNN-STD	MTL-50 FNN-STD	MTL-75 FNN-STD	TO XGB-STD	NP XGB-STD	MTL-25 XGB-STD	MTL-50 XGB-STD	MTL-75 XGB-STD
RW	–	0.807	0.745*	0.723*	0.719*	0.717*	0.845	0.721*	0.729*	0.724*	0.720*	1.127	0.770*	0.789*	0.764*	0.769*
TO HAR-STD	1.336	–	0.977	0.934	0.930	0.927	1.042	0.943	0.939	0.934	0.929	1.402	0.968	0.984	0.963	0.958
TO HAR-EXT	1.249	0.873	0.898	0.849	0.845	0.841	0.908	0.865	0.853	0.847	0.842	1.259	0.856	0.867	0.856	0.840
NP HAR-STD	1.353	1.063	–	0.964	0.960	0.956	1.112	0.967	0.971	0.965	0.960	1.480	1.015	1.039	1.009	1.012
NP HAR-EXT	1.378	1.049	1.007	0.967	0.962	0.959	1.099	0.974	0.974	0.967	0.962	1.468	1.003	1.028	1.001	0.998
MTL-25 HAR-STD	1.415	1.094	1.042	–	0.996	0.993	1.143	1.006	1.007	1.001	0.996	1.521	1.046	1.070	1.041	1.042
MTL-50 HAR-STD	1.421	1.099	1.046	1.004	–	0.997	1.148	1.010	1.012	1.005	1.000	1.528	1.050	1.074	1.046	1.046
MTL-75 HAR-STD	1.425	1.102	1.049	1.007	1.003	–	1.151	1.013	1.015	1.008	1.003	1.533	1.054	1.078	1.049	1.050
MTL-25 HAR-EXT	1.521	1.151	1.112	1.063	1.058	1.055	1.203	1.073	1.070	1.063	1.058	1.594	1.101	1.124	1.098	1.094
MTL-50 HAR-EXT	1.526	1.153	1.115	1.066	1.062	1.058	1.205	1.077	1.073	1.066	1.061	1.598	1.103	1.127	1.100	1.096
MTL-75 HAR-EXT	1.537	1.160	1.123	1.073	1.069	1.065	1.213	1.084	1.080	1.073	1.068	1.612	1.110	1.134	1.108	1.103
TO FNN-STD	1.288	0.961	0.942	0.899	0.896	0.893*	–	0.908	0.905	0.899	0.895*	1.352	0.932	0.948	0.927	0.923*
TO FNN-EXT	1.307	0.921	0.942	0.891	0.887	0.884	0.957	0.907	0.895	0.890	0.885	1.310	0.902	0.913	0.901	0.887
NP FNN-STD	1.401	1.097	1.035	0.997	0.992	0.989	1.147	–	1.004	0.998	0.993	1.527	1.048	1.073	1.042	1.045
NP FNN-EXT	1.258	0.974	0.922	0.890	0.885	0.882	1.024	0.894	0.897	0.891	0.885	1.375	0.928	0.952	0.925	0.924
MTL-25 FNN-STD	1.407	1.084	1.035	0.993	0.989	0.986	1.133	0.999	–	0.994	0.989	1.508	1.037	1.060	1.033	1.033
MTL-50 FNN-STD	1.415	1.091	1.041	0.999	0.995	0.992	1.140	1.005	1.006	–	0.995	1.518	1.044	1.068	1.040	1.040
MTL-75 FNN-STD	1.422	1.097	1.046	1.004	1.000	0.997	1.146	1.010	1.011	1.005	–	1.527	1.050	1.073	1.045	1.045
MTL-25 FNN-EXT	1.509	1.143	1.103	1.055	1.051	1.047	1.195	1.065	1.062	1.055	1.050	1.586	1.093	1.117	1.090	1.087
MTL-50 FNN-EXT	1.531	1.162	1.120	1.071	1.067	1.063	1.215	1.081	1.079	1.072	1.066	1.613	1.111	1.135	1.108	1.105
MTL-75 FNN-EXT	1.570	1.199	1.150	1.101	1.096	1.093	1.255	1.110	1.109	1.102	1.096	1.665	1.144	1.171	1.141	1.138
TO XGB-STD	0.991	0.742*	0.723*	0.689*	0.686*	0.684*	0.776*	0.697*	0.693*	0.689*	0.686*	–	0.715*	0.725*	0.710*	0.707*
TO XGB-EXT	1.062	0.778	0.773	0.737*	0.733*	0.731*	0.813	0.745	0.741*	0.736*	0.732*	1.065	0.764*	0.768*	0.755*	0.749*
NP XGB-STD	1.391	1.049	1.016	0.970	0.966	0.963	1.095	0.980	0.976	0.970	0.966	1.464	–	1.026	1.001	0.997
NP XGB-EXT	1.465	1.109	1.068	1.020	1.016	1.012	1.161	1.031	1.027	1.020	1.015	1.545	1.049	1.080	1.053	1.048
MTL-25 XGB-STD	1.359	1.021	0.993	0.948	0.944	0.941	1.066	0.958	0.954	0.948	0.944	1.419	0.981	–	0.978	0.974
MTL-50 XGB-STD	1.384	1.047	1.013	0.968	0.964	0.961	1.093	0.977	0.974	0.968	0.964	1.455	1.003	1.024	–	0.997
MTL-75 XGB-STD	1.400	1.050	1.022	0.975	0.971	0.968	1.095	0.986	0.981	0.975	0.971	1.462	1.007	1.028	1.005	–
MTL-25 XGB-EXT	1.552	1.166	1.131	1.080	1.075	1.071	1.218	1.091	1.087	1.080	1.074	1.619	1.115	1.140	1.112	1.107
MTL-50 XGB-EXT	1.555	1.174	1.134	1.084	1.079	1.075	1.228	1.094	1.092	1.084	1.079	1.628	1.121	1.146	1.118	1.114
MTL-75 XGB-EXT	1.618	1.216	1.178	1.126	1.121	1.117	1.271	1.137	1.134	1.126	1.120	1.695	1.162	1.189	1.160	1.154

	TO HAR-EXT	NP HAR-EXT	MTL-25 HAR-EXT	MTL-50 HAR-EXT	MTL-75 HAR-EXT	TO FNN-EXT	NP FNN-EXT	MTL-25 FNN-EXT	MTL-50 FNN-EXT	MTL-75 FNN-EXT	TO XGB-EXT	NP XGB-EXT	MTL-25 XGB-EXT	MTL-50 XGB-EXT	MTL-75 XGB-EXT
RW	21.310	0.782*	0.698*	0.698*	0.695*	2.939	0.916	0.702*	0.690*	0.668*	1.198	0.751*	0.699*	0.691*	0.673*
TO HAR-STD	13.402	1.002	0.881*	0.879*	0.874*	2.282	1.204	0.888*	0.874*	0.854*	1.388	0.951*	0.877*	0.872*	0.846*
TO HAR-EXT	–	0.873	0.778*	0.773*	0.765*	0.861	1.059	0.784*	0.773*	0.757*	1.120	0.835*	0.762*	0.764	0.733*
NP HAR-STD	23.633	1.038	0.922*	0.922*	0.917*	3.405	1.226	0.928*	0.912*	0.886*	1.543	0.990	0.921	0.911*	0.887*
NP HAR-EXT	18.259	–	0.910	0.907	0.900	2.817	1.170	0.916	0.901	0.874	1.462	0.969	0.897	0.893	0.862
MTL-25 HAR-STD	21.317	1.077	0.951	0.950	0.946	3.191	1.285	0.958	0.942*	0.917	1.566	1.021	0.949	0.940	0.915*
MTL-50 HAR-STD	21.499	1.081	0.955	0.954	0.950	3.214	1.289	0.962	0.946*	0.921	1.572	1.025	0.953	0.944	0.918*
MTL-75 HAR-STD	21.574	1.085	0.959	0.958	0.953	3.224	1.294	0.965	0.949*	0.923	1.577	1.028	0.956	0.947	0.921
MTL-25 HAR-EXT	18.121	1.135	–	0.998	0.993	2.904	1.363	1.007	0.992	0.966	1.602	1.069	0.995	0.987	0.959
MTL-50 HAR-EXT	17.763	1.137	1.002	–	0.995	2.867	1.366	1.009	0.994	0.968	1.600	1.071	0.996	0.989	0.960
MTL-75 HAR-EXT	17.859	1.140	1.008	1.006	–	2.882	1.370	1.015	1.000	0.973	1.608	1.076	1.001	0.994	0.964
TO FNN-STD	12.542	0.968	0.849*	0.847*	0.843*	2.159	1.169	0.856*	0.843*	0.823*	1.337	0.917*	0.846*	0.841*	0.816*
TO FNN-EXT	1.869	0.930	0.821*	0.817*	0.811	–	1.129	0.829*	0.818	0.802	1.192	0.885	0.810	0.811	0.781*
NP FNN-STD	23.788	1.075	0.953*	0.952*	0.948*	3.452	1.276	0.959	0.943*	0.916*	1.587	1.023	0.951	0.941	0.916*
NP FNN-EXT	19.305	0.904	0.840	0.837	0.830	2.855	–	0.845	0.830	0.803	1.374	0.896	0.826	0.820	0.792
MTL-25 FNN-STD	20.482	1.070	0.944	0.943	0.938	3.095	1.277	0.950	0.935*	0.910	1.547	1.013	0.942	0.933	0.908*
MTL-50 FNN-STD	20.839	1.076	0.950	0.949	0.944	3.139	1.284	0.956	0.941*	0.916	1.558	1.019	0.948	0.939	0.913*
MTL-75 FNN-STD	21.096	1.081	0.955	0.954	0.949	3.170	1.290	0.961	0.945*	0.920	1.567	1.024	0.952	0.943	0.917
MTL-25 FNN-EXT	18.363	1.127	0.993	0.991	0.986	2.922	1.352	–	0.984	0.959	1.595	1.061	0.987	0.980	0.951
MTL-50 FNN-EXT	19.246	1.144	1.009	1.007	1.002	3.030	1.372	1.016	–	0.974	1.625	1.077	1.003	0.995	0.966
MTL-75 FNN-EXT	21.313	1.172	1.038	1.036	1.030	3.276	1.401	1.045	1.028	–	1.684	1.105	1.031	1.023	0.992
TO XGB-STD	10.240	0.742	0.648*	0.647*	0.644*	1.727	0.903	0.653*	0.644*	0.629*	1.008	0.702*	0.645*	0.641*	0.624*
TO XGB-EXT	6.394	0.774	0.685*	0.683*	0.678*	1.352	0.943	0.692*	0.681*	0.666*	–	0.749*	0.678*	0.678*	0.655*
NP XGB-STD	16.410	1.038	0.914	0.913	0.908	2.636	1.245	0.921	0.907	0.884	1.479	0.973	0.910	0.903	0.877
NP XGB-EXT	18.731	1.079	0.956	0.954	0.948	2.923	1.298	0.963	0.947	0.920	1.554	–	0.948	0.940	0.910
MTL-25 XGB-STD	14.710	1.019	0.893	0.891	0.887	2.437	1.225	0.900	0.886	0.865	1.421	0.959	0.889*	0.883	0.858*
MTL-50 XGB-STD	16.577	1.040	0.914	0.912	0.908	2.653	1.247	0.920	0.906	0.883	1.468	0.980	0.910	0.902	0.877*
MTL-75 XGB-STD	15.133	1.045	0.918	0.916	0.911	2.504	1.256	0.924	0.910	0.888	1.462	0.982	0.912	0.906	0.880
MTL-25 XGB-EXT	17.920	1.139	1.012	1.009	1.002	2.889	1.364	1.019	1.003	0.976	1.609	1.077	–	0.994	0.962
MTL-50 XGB-EXT	19.200	1.149	1.018	1.016	1.009	3.032	1.372	1.025	1.008	0.981	1.633	1.083	1.008	–	0.970
MTL-75 XGB-EXT	19.063	1.183	1.055	1.051	1.044	3.049	1.413	1.061	1.044	1.015	1.680	1.118	1.040	1.034	–

Table E.11
One-day-ahead cross-sectional average relative QLIKE losses. Each value in this table represents the cross-sectional average of the pairwise realized variance forecast QLIKE loss for the model in the selected column relative to the benchmark in the selected row. Each model's QLIKE loss is determined for one-day-ahead realized variance forecasts over 500 days, starting 50 days after the first trading day of each target asset. The symbol (∗) denotes that the Diebold–Mariano test of equal predictive accuracy is rejected for more than 50% of the target assets using a 5% significance level. A rejection for the majority of target assets indicates that the model in the column exhibits a significantly lower QLIKE loss than the corresponding benchmark model in the respective row. The corresponding QLIKE loss values are additionally marked in blue. '—' means not applicable. Model names reflect both the forecasting approach and predictor set: TO denotes models trained solely on the target data; NP denotes naive pooling, where models are trained on both target and all available source data; and MTL denotes multisource transfer learning, where the numerical suffix (25, 50, 75) indicates the percentage of subsequences transferred from the source data into the final training data set. The second part of the name indicates the base model class, with HAR referring to the heterogeneous autoregressive model, FNN to feedforward neural networks, and XGB to XGBoost, while STD refers to the standard HAR predictor set, and EXT to an extended predictor set.

	RW	TO HAR-STD	NP HAR-STD	MTL-25 HAR-STD	MTL-50 HAR-STD	MTL-75 HAR-STD	TO FNN-STD	NP FNN-STD	MTL-25 FNN-STD	MTL-50 FNN-STD	MTL-75 FNN-STD	TO XGB-STD	NP XGB-STD	MTL-25 XGB-STD	MTL-50 XGB-STD	MTL-75 XGB-STD
RW	–	0.854	0.773*	0.817*	0.808*	0.798*	0.951	0.768*	0.819*	0.809*	0.799*	1.164	0.806*	0.907*	0.811*	0.793*
TO HAR-STD	1.185	–	0.913*	0.961	0.951	0.939	1.122	0.905*	0.963	0.952	0.939	1.341	0.952*	1.063	0.954	0.934*
TO HAR-EXT	0.821	0.684	0.627*	0.659	0.651	0.641	0.756	0.622*	0.660	0.652	0.642	0.882	0.638	0.709	0.647	0.633*
NP HAR-STD	1.298	1.104	–	1.056	1.045	1.032	1.229	0.994	1.059	1.046	1.033	1.498	1.041	1.176	1.048	1.026
NP HAR-EXT	1.161	0.997	0.900	0.953	0.943	0.932	1.118	0.895	0.956	0.945	0.932	1.358	0.943	1.057	0.945	0.925
MTL-25 HAR-STD	1.233	1.045	0.949	–	0.990*	0.978*	1.164	0.942*	1.003	0.991*	0.978*	1.415	0.988	1.112	0.993	0.973
MTL-50 HAR-STD	1.245	1.055	0.958	1.010	–	0.987	1.176	0.951*	1.013	1.001	0.988	1.426	0.998	1.123	1.003	0.982
MTL-75 HAR-STD	1.261	1.069	0.971	1.023	1.013	–	1.190	0.964	1.026	1.014	1.001	1.443	1.010	1.137	1.016	0.995
MTL-25 HAR-EXT	1.378	1.173	1.061	1.121	1.109	1.095	1.299	1.055	1.124	1.110	1.096	1.575	1.100	1.240	1.109	1.086
MTL-50 HAR-EXT	1.386	1.179	1.068	1.128	1.116	1.101	1.307	1.062	1.131	1.117	1.102	1.578	1.106	1.247	1.116	1.092
MTL-75 HAR-EXT	1.407	1.198	1.085	1.146	1.134	1.119	1.326	1.078	1.149	1.135	1.119	1.599	1.122	1.266	1.133	1.108
TO FNN-STD	1.076	0.912*	0.827*	0.872*	0.863*	0.852*	–	0.822*	0.875*	0.864*	0.852*	1.221	0.854*	0.974	0.864*	0.845*
TO FNN-EXT	1.052	0.877	0.809	0.853	0.842	0.830	0.987	0.801	0.854	0.843	0.830	1.091	0.833	0.926	0.839	0.819
NP FNN-STD	1.307	1.110	1.007	1.062	1.052	1.038	1.238	–	1.065	1.053	1.039	1.502	1.050	1.181	1.055	1.033
NP FNN-EXT	1.073	0.925*	0.834*	0.884	0.875*	0.863*	1.037	0.830*	0.887*	0.876*	0.863*	1.269	0.872*	0.968	0.874	0.855*
MTL-25 FNN-STD	1.229	1.042	0.946	0.997	0.987	0.975*	1.162	0.939*	–	0.988*	0.976*	1.409	0.985	1.109	0.991	0.970
MTL-50 FNN-STD	1.243	1.054	0.957	1.009	0.999	0.986*	1.175	0.950*	1.012	–	0.987*	1.424	0.997	1.122	1.002	0.981
MTL-75 FNN-STD	1.260	1.068	0.970	1.023	1.012	0.999	1.189	0.963	1.025	1.013	–	1.442	1.010	1.136	1.015	0.994
MTL-25 FNN-EXT	1.375	1.171	1.059	1.119	1.107	1.093	1.297	1.053	1.122	1.109	1.094	1.579	1.098	1.236	1.108	1.084
MTL-50 FNN-EXT	1.396	1.190	1.076	1.137	1.125	1.110	1.318	1.070	1.140	1.126	1.111	1.603	1.116	1.257	1.125	1.101
MTL-75 FNN-EXT	1.434	1.223	1.106	1.168	1.156	1.141	1.354	1.100	1.172	1.158	1.142	1.648	1.146	1.293	1.156	1.131
TO XGB-STD	0.932	0.781*	0.716*	0.754*	0.745*	0.735*	0.874	0.709*	0.755*	0.746*	0.736*	–	0.740*	0.836*	0.745*	0.729*
TO XGB-EXT	1.154	0.977	0.891*	0.941	0.930	0.917	1.099	0.884*	0.943	0.931	0.918	1.290	0.927*	1.033	0.930	0.910
NP XGB-STD	1.259	1.071	0.968	1.022	1.012	0.999	1.179	0.963	1.025	1.013	0.999	1.430	–	1.142	1.013	0.991
NP XGB-EXT	1.336	1.140	1.029	1.088	1.076	1.062	1.251	1.024	1.091	1.078	1.063	1.514	1.059	1.205	1.074	1.050
MTL-25 XGB-STD	1.152	0.973	0.888	0.935	0.925	0.913	1.091	0.881	0.937	0.926	0.914	1.314	0.925	–	0.926	0.907
MTL-50 XGB-STD	1.242	1.053	0.956	1.008	0.998	0.985	1.171	0.950	1.011	0.999	0.986	1.417	0.994	1.117	–	0.979
MTL-75 XGB-STD	1.269	1.077	0.977	1.031	1.020	1.007	1.196	0.971	1.034	1.021	1.008	1.444	1.014	1.143	1.022	–
MTL-25 XGB-EXT	1.385	1.179	1.069	1.129	1.117	1.102	1.309	1.062	1.132	1.118	1.103	1.578	1.108	1.245	1.116	1.092
MTL-50 XGB-EXT	1.422	1.212	1.098	1.160	1.147	1.132	1.346	1.091	1.163	1.149	1.133	1.626	1.139	1.277	1.147	1.122
MTL-75 XGB-EXT	1.469	1.255	1.135	1.200	1.187	1.171	1.388	1.129	1.204	1.189	1.172	1.685	1.175	1.324	1.186	1.160

	TO HAR-EXT	NP HAR-EXT	MTL-25 HAR-EXT	MTL-50 HAR-EXT	MTL-75 HAR-EXT	TO FNN-EXT	NP FNN-EXT	MTL-25 FNN-EXT	MTL-50 FNN-EXT	MTL-75 FNN-EXT	TO XGB-EXT	NP XGB-EXT	MTL-25 XGB-EXT	MTL-50 XGB-EXT	MTL-75 XGB-EXT
RW	1.706	0.887*	0.734*	0.731*	0.721*	1.337	0.997	0.736*	0.724*	0.706*	0.896	0.771*	0.733*	0.713*	0.691*
TO HAR-STD	1.961	1.059	0.867*	0.863*	0.851*	1.482	1.197	0.871*	0.856*	0.835*	1.050	0.913*	0.865	0.842*	0.819*
TO HAR-EXT	–	0.754	0.581*	0.576*	0.566*	0.876	0.840	0.583*	0.572*	0.557*	0.693	0.596*	0.578*	0.563*	0.546*
NP HAR-STD	2.179	1.157	0.950	0.945	0.933	1.709	1.305	0.953	0.938	0.914	1.160	0.997	0.949	0.924	0.896
NP HAR-EXT	2.070	–	0.852*	0.848*	0.837*	1.581	1.122	0.854	0.840*	0.819*	1.035	0.899*	0.847*	0.824*	0.800*
MTL-25 HAR-STD	2.058	1.101	0.901*	0.896*	0.885*	1.605	1.244	0.904	0.889*	0.867*	1.100	0.946	0.900	0.876*	0.851*
MTL-50 HAR-STD	2.071	1.112	0.910	0.905*	0.894*	1.611	1.256	0.913	0.898	0.876*	1.110	0.955	0.909	0.885	0.859*
MTL-75 HAR-STD	2.089	1.127	0.922	0.917	0.905	1.626	1.271	0.925	0.910	0.887	1.123	0.967	0.920	0.896	0.870*
MTL-25 HAR-EXT	2.280	1.224	–	0.995	0.982	1.769	1.380	1.003	0.987	0.962	1.220	1.045	0.997	0.971	0.942
MTL-50 HAR-EXT	2.274	1.232	1.006	–	0.986	1.759	1.387	1.008	0.992	0.967	1.224	1.050	1.002	0.976	0.947
MTL-75 HAR-EXT	2.285	1.250	1.020	1.014	–	1.771	1.405	1.023	1.007	0.981	1.240	1.064	1.016	0.990	0.960
TO FNN-STD	1.759	0.969	0.783*	0.779*	0.769*	1.357	1.097	0.786*	0.773*	0.754*	0.962	0.815*	0.783*	0.763*	0.739*
TO FNN-EXT	1.592	0.940	0.758*	0.751*	0.738*	–	1.062	0.763*	0.749*	0.730*	0.892	0.789*	0.750*	0.732*	0.712*
NP FNN-STD	2.187	1.166	0.957	0.952	0.940	1.702	1.317	0.960	0.944	0.921	1.166	1.005	0.955	0.930	0.903
NP FNN-EXT	1.778	0.921	0.786	0.781*	0.769*	1.475	–	0.786	0.773*	0.751*	0.949	0.823*	0.780*	0.758*	0.734*
MTL-25 FNN-STD	2.049	1.098	0.899*	0.894*	0.883*	1.596	1.241	0.902	0.887*	0.865*	1.096	0.944	0.898	0.874	0.849*
MTL-50 FNN-STD	2.068	1.111	0.909	0.904*	0.893*	1.608	1.255	0.912	0.897	0.875*	1.108	0.954	0.908	0.884	0.858*
MTL-75 FNN-STD	2.087	1.126	0.921	0.916	0.904	1.624	1.271	0.924	0.909	0.886	1.122	0.966	0.920	0.895	0.869*
MTL-25 FNN-EXT	2.271	1.221	0.998	0.993	0.979	1.787	1.374	–	0.985	0.960	1.218	1.042	0.995	0.969	0.940
MTL-50 FNN-EXT	2.284	1.240	1.014	1.008	0.995	1.810	1.394	1.016	–	0.974	1.235	1.059	1.011	0.984	0.954
MTL-75 FNN-EXT	2.333	1.273	1.041	1.035	1.021	1.860	1.428	1.043	1.027	–	1.268	1.086	1.037	1.010	0.979
TO XGB-STD	1.529	0.830	0.676*	0.671*	0.662*	1.008	0.945	0.679*	0.668*	0.652*	0.812	0.707*	0.673*	0.656*	0.638*
TO XGB-EXT	1.830	1.018	0.838*	0.832*	0.819*	1.383	1.142	0.841*	0.826*	0.805*	–	0.878*	0.831*	0.809*	0.786*
NP XGB-STD	2.056	1.127	0.916	0.911	0.899	1.588	1.273	0.919	0.904	0.881	1.121	0.953	0.915	0.892	0.864
NP XGB-EXT	2.124	1.190	0.965	0.959	0.945	1.667	1.336	0.968	0.952	0.927	1.177	–	0.963	0.938	0.907
MTL-25 XGB-STD	1.906	1.029	0.839*	0.835*	0.824*	1.463	1.157	0.841*	0.828*	0.808*	1.020	0.882	0.837	0.814*	0.792*
MTL-50 XGB-STD	2.053	1.108	0.906	0.901	0.889	1.589	1.251	0.909	0.894	0.872*	1.104	0.949	0.904	0.880*	0.855*
MTL-75 XGB-STD	2.098	1.132	0.925	0.920	0.908	1.610	1.276	0.928	0.914	0.891	1.128	0.969	0.924	0.899	0.873
MTL-25 XGB-EXT	2.283	1.226	1.005	0.999	0.985	1.758	1.380	1.008	0.991	0.966	1.219	1.050	–	0.974	0.944
MTL-50 XGB-EXT	2.339	1.256	1.032	1.026	1.012	1.819	1.413	1.035	1.018	0.992	1.253	1.079	1.027	–	0.970
MTL-75 XGB-EXT	2.403	1.299	1.066	1.060	1.045	1.890	1.456	1.069	1.051	1.024	1.295	1.112	1.061	1.033	–

Table F.12
Average selection rates of source asset subsequences for MTL-25 models of individual target assets. Source assets that belong to the same GICS sector as the respective target asset are highlighted in gray. Selection rates greater than 45% are highlighted in blue, while selection rates of less than 5% are highlighted in red. Dashes (−) indicate the absence of available subsequences for selection, due to the later distribution date of the source asset compared to the target asset.

GICS sector	Source asset	Target asset									
		TWTR	NCLH	LW	PSX	SYF	MRNA	CARR	DXC	CTVA	INVH
Communication Services	DISH	0.524	0.415	0.248	0.392	0.374	0.521	0.384	0.227	0.439	0.100
	EA	0.620	0.328	0.177	0.223	0.310	0.572	0.358	0.182	0.382	0.094
	GOOGL	0.140	0.189	0.229	0.172	0.217	0.198	0.170	0.229	0.154	0.250
	LUMN	0.042	0.130	0.290	0.137	0.168	0.241	0.242	0.292	0.223	0.323
	META	0.658	0.379	0.193	0.284	0.337	0.409	0.303	0.190	0.301	0.134
	TTWO	0.697	0.294	0.146	0.139	0.296	0.687	0.395	0.158	0.418	0.054
Consumer Discretionary	AMZN	0.429	0.412	0.211	0.412	0.365	0.350	0.260	0.182	0.291	0.105
	EBAY	0.400	0.479	0.365	0.515	0.426	0.228	0.363	0.309	0.377	0.242
	LEN	0.707	0.309	0.208	0.145	0.297	0.595	0.366	0.206	0.390	0.113
	MHK	0.584	0.455	0.284	0.381	0.399	0.401	0.360	0.257	0.391	0.187
	NWL	0.233	0.346	0.338	0.431	0.328	0.323	0.302	0.304	0.300	0.291
	TSLA	0.487	0.148	0.096	0.061	0.190	0.868	0.335	0.215	0.328	0.088
Consumer Staples	CPB	0.026	0.097	0.222	0.106	0.102	0.052	0.159	0.233	0.142	0.291
	HSY	0.032	0.086	0.185	0.107	0.090	0.024	0.105	0.221	0.084	0.288
	KMB	0.022	0.065	0.133	0.077	0.064	0.017	0.091	0.162	0.072	0.217
	PG	0.021	0.047	0.076	0.059	0.046	0.017	0.052	0.116	0.037	0.166
	TAP	0.064	0.156	0.280	0.142	0.171	0.091	0.197	0.273	0.167	0.319
	WMT	0.024	0.055	0.120	0.054	0.055	0.025	0.066	0.156	0.055	0.225
Energy	APA	0.306	0.366	0.268	0.462	0.357	0.527	0.339	0.257	0.387	0.189
	CVX	0.102	0.159	0.223	0.190	0.191	0.104	0.194	0.250	0.170	0.316
	DVN	0.313	0.390	0.270	0.456	0.365	0.508	0.338	0.243	0.389	0.161
	EQT	0.321	0.437	0.313	0.452	0.423	0.438	0.348	0.255	0.418	0.146
	HES	0.381	0.395	0.246	0.481	0.368	0.548	0.388	0.233	0.438	0.139
	XOM	0.060	0.112	0.176	0.125	0.135	0.071	0.141	0.201	0.114	0.264
Financials	AIZ	0.136	0.253	0.354	0.306	0.276	0.114	0.244	0.347	0.224	0.380
	BRK-B	0.078	0.090	0.091	0.111	0.084	0.051	0.078	0.110	0.076	0.141
	JPM	0.225	0.337	0.310	0.440	0.327	0.135	0.248	0.308	0.240	0.337
	LNC	0.455	0.415	0.321	0.367	0.381	0.337	0.375	0.280	0.413	0.194
	MTB	0.194	0.212	0.310	0.230	0.218	0.155	0.249	0.327	0.219	0.370
	NDAQ	0.236	0.374	0.350	0.423	0.377	0.140	0.261	0.314	0.255	0.316
Health Care	BIIB	0.405	0.357	0.289	0.342	0.371	0.424	0.384	0.257	0.416	0.167
	DVA	0.069	0.163	0.309	0.167	0.181	0.136	0.224	0.315	0.194	0.375
	DXCM	0.630	0.353	0.056	0.147	0.263	0.929	0.297	0.203	0.282	0.025
	JNJ	0.025	0.045	0.079	0.055	0.050	0.033	0.058	0.102	0.048	0.136
	UNH	0.218	0.288	0.294	0.325	0.286	0.157	0.248	0.289	0.240	0.330
	XRAY	0.077	0.185	0.350	0.207	0.214	0.105	0.220	0.340	0.186	0.409
Industrials	ALK	0.677	0.480	0.252	0.398	0.418	0.477	0.456	0.244	0.542	0.084
	FAST	0.240	0.402	0.387	0.431	0.394	0.172	0.326	0.323	0.354	0.284
	GNRC	0.509	0.371	0.201	0.241	0.326	0.616	0.347	0.233	0.401	0.092
	GWW	0.166	0.281	0.363	0.291	0.313	0.239	0.328	0.318	0.321	0.298
	HON	0.121	0.195	0.221	0.256	0.195	0.077	0.151	0.239	0.138	0.287
	UPS	0.082	0.117	0.145	0.152	0.114	0.072	0.142	0.181	0.124	0.234
Information Technology	AAPL	0.178	0.278	0.311	0.270	0.282	0.123	0.232	0.288	0.221	0.302
	FFIV	0.670	0.361	0.224	0.223	0.333	0.465	0.369	0.218	0.388	0.138
	FTNT	0.655	0.291	0.169	0.107	0.254	0.618	0.345	0.174	0.384	0.074
	MSFT	0.110	0.258	0.342	0.262	0.297	0.112	0.214	0.328	0.189	0.379
	QRVO	0.696	0.416	0.145	–	0.279	0.671	0.307	0.214	0.346	0.080
	TEL	0.195	0.301	0.304	0.385	0.305	0.126	0.234	0.299	0.215	0.346
Materials	APD	0.125	0.167	0.245	0.167	0.176	0.083	0.168	0.261	0.149	0.323
	IFF	0.101	0.161	0.300	0.183	0.186	0.103	0.200	0.302	0.177	0.358
	LIN	–	–	0.733	–	–	0.136	0.192	0.558	0.191	0.423
	SEE	0.314	0.384	0.350	0.388	0.394	0.216	0.327	0.307	0.320	0.277
	VMC	0.544	0.413	0.252	0.354	0.371	0.432	0.403	0.231	0.446	0.116
	WRK	0.809	0.938	0.284	–	0.574	0.390	0.383	0.270	0.407	0.197
Real Estate	AMT	0.044	0.126	0.261	0.199	0.129	0.030	0.167	0.294	0.133	0.399
	AVB	0.140	0.223	0.354	0.244	0.254	0.089	0.218	0.367	0.175	0.465
	FRT	0.143	0.192	0.344	0.199	0.226	0.110	0.226	0.369	0.186	0.450
	PLD	0.261	0.329	0.329	0.397	0.317	0.135	0.254	0.330	0.234	0.384
	VNO	0.034	0.089	0.255	0.105	0.108	0.052	0.144	0.299	0.098	0.400
	WY	0.314	0.394	0.353	0.476	0.357	0.191	0.313	0.327	0.304	0.333
Utilities	AWK	0.056	0.184	0.355	0.216	0.218	0.027	0.188	0.399	0.137	0.534
	DUK	0.031	0.124	0.186	0.115	0.135	0.023	0.105	0.260	0.077	0.342
	ES	0.036	0.125	0.246	0.140	0.148	0.024	0.146	0.298	0.107	0.398
	NEE	0.036	0.121	0.227	0.138	0.138	0.023	0.128	0.279	0.093	0.388
	NRG	0.385	0.443	0.279	0.482	0.414	0.426	0.401	0.259	0.428	0.174
	PNW	0.030	0.094	0.233	0.101	0.129	0.029	0.150	0.260	0.102	0.356

Table F.13
Average selection rates of source asset subsequences for MTL-50 models of individual target assets. Source assets that belong to the same GICS sector as the respective target asset are highlighted in gray. Selection rates greater than 70% are highlighted in blue, while selection rates of less than 30% are highlighted in red. Dashes (–) indicate the absence of available subsequences for selection, due to the later distribution date of the source asset compared to the target asset.

GICS sector	Source asset	Target asset									
		TWTR	NCLH	LW	PSX	SYF	MRNA	CARR	DXC	CTVA	INVH
Communication Services	DISH	0.863	0.610	0.388	0.536	0.556	0.926	0.591	0.345	0.655	0.179
	EA	0.834	0.475	0.311	0.353	0.452	0.881	0.572	0.310	0.603	0.182
	GOOGL	0.416	0.493	0.497	0.518	0.501	0.431	0.442	0.479	0.439	0.503
	LUMN	0.208	0.420	0.593	0.418	0.490	0.454	0.499	0.590	0.495	0.617
	META	0.861	0.521	0.383	0.330	0.510	0.690	0.561	0.385	0.571	0.317
	TTWO	0.831	0.380	0.234	0.229	0.394	0.956	0.566	0.248	0.593	0.104
Consumer Discretionary	AMZN	0.853	0.641	0.373	0.570	0.579	0.730	0.501	0.347	0.531	0.243
	EBAY	0.848	0.747	0.600	0.697	0.693	0.656	0.670	0.550	0.710	0.451
	LEN	0.839	0.418	0.327	0.217	0.424	0.882	0.588	0.335	0.627	0.202
	MHK	0.883	0.657	0.474	0.517	0.607	0.766	0.607	0.442	0.659	0.325
	NWL	0.647	0.697	0.603	0.705	0.665	0.625	0.592	0.566	0.601	0.520
	TSLA	0.559	0.188	0.141	0.070	0.219	0.968	0.349	0.250	0.397	0.067
Consumer Staples	CPB	0.093	0.295	0.571	0.282	0.337	0.217	0.421	0.601	0.382	0.732
	HSY	0.135	0.274	0.531	0.336	0.307	0.141	0.367	0.574	0.314	0.725
	KMB	0.075	0.206	0.429	0.232	0.230	0.105	0.303	0.501	0.255	0.661
	PG	0.059	0.159	0.309	0.186	0.165	0.051	0.185	0.398	0.144	0.572
	TAP	0.268	0.456	0.617	0.466	0.502	0.358	0.508	0.607	0.478	0.661
	WMT	0.061	0.184	0.419	0.217	0.218	0.098	0.254	0.494	0.203	0.665
Energy	APA	0.670	0.685	0.456	0.725	0.634	0.800	0.527	0.435	0.602	0.327
	CVX	0.259	0.390	0.532	0.495	0.426	0.272	0.414	0.555	0.376	0.657
	DVN	0.755	0.717	0.434	0.730	0.661	0.839	0.553	0.405	0.620	0.280
	EQT	0.844	0.734	0.490	0.710	0.685	0.902	0.574	0.433	0.658	0.275
	HES	0.770	0.692	0.409	0.671	0.620	0.860	0.588	0.389	0.648	0.257
	XOM	0.196	0.321	0.473	0.394	0.367	0.203	0.348	0.519	0.290	0.652
Financials	AIZ	0.434	0.620	0.678	0.736	0.628	0.408	0.590	0.650	0.592	0.655
	BRK-B	0.163	0.235	0.304	0.309	0.232	0.140	0.229	0.390	0.202	0.526
	JPM	0.606	0.665	0.615	0.732	0.642	0.408	0.544	0.602	0.521	0.631
	LNC	0.829	0.645	0.521	0.531	0.616	0.779	0.668	0.483	0.721	0.354
	MTB	0.389	0.473	0.616	0.563	0.488	0.389	0.538	0.617	0.524	0.656
	NDAQ	0.738	0.736	0.651	0.724	0.713	0.473	0.607	0.608	0.598	0.608
Health Care	BIIB	0.786	0.643	0.470	0.677	0.621	0.844	0.658	0.440	0.704	0.314
	DVA	0.277	0.515	0.662	0.572	0.537	0.363	0.508	0.648	0.495	0.706
	DXCM	0.708	0.404	0.087	0.216	0.308	0.978	0.337	0.200	0.356	0.029
	JNJ	0.076	0.144	0.282	0.169	0.163	0.103	0.207	0.376	0.164	0.523
	UNH	0.540	0.629	0.597	0.683	0.623	0.423	0.556	0.574	0.544	0.598
	XRAY	0.297	0.540	0.707	0.635	0.584	0.336	0.537	0.682	0.525	0.736
Industrials	ALK	0.933	0.630	0.379	0.516	0.569	0.949	0.662	0.345	0.745	0.145
	FAST	0.751	0.746	0.648	0.742	0.710	0.654	0.705	0.588	0.734	0.495
	GNRC	0.854	0.547	0.278	0.325	0.525	0.977	0.528	0.278	0.595	0.116
	GWW	0.524	0.655	0.640	0.666	0.662	0.621	0.662	0.596	0.690	0.534
	HON	0.322	0.483	0.519	0.623	0.488	0.231	0.387	0.547	0.347	0.652
	UPS	0.191	0.308	0.440	0.384	0.325	0.204	0.363	0.503	0.307	0.638
Information Technology	AAPL	0.510	0.648	0.627	0.664	0.650	0.389	0.539	0.600	0.538	0.607
	FFIV	0.871	0.500	0.383	0.321	0.493	0.802	0.635	0.378	0.672	0.248
	FTNT	0.801	0.371	0.268	0.160	0.366	0.938	0.555	0.277	0.598	0.133
	MSFT	0.432	0.642	0.670	0.711	0.676	0.306	0.523	0.646	0.522	0.687
	QRVO	0.734	0.459	0.185	–	0.356	0.945	0.452	0.244	0.494	0.081
	TEL	0.541	0.631	0.616	0.714	0.617	0.375	0.525	0.601	0.496	0.640
Materials	APD	0.296	0.458	0.573	0.499	0.489	0.244	0.443	0.577	0.405	0.674
	IFF	0.302	0.467	0.623	0.513	0.513	0.357	0.510	0.614	0.485	0.653
	LIN	–	–	0.689	–	–	0.245	0.484	0.748	0.381	0.727
	SEE	0.757	0.696	0.606	0.668	0.671	0.590	0.649	0.557	0.669	0.487
	VMC	0.884	0.618	0.415	0.492	0.575	0.895	0.684	0.384	0.729	0.211
	WRK	0.882	0.938	0.480	–	0.591	0.753	0.626	0.448	0.680	0.320
Real Estate	AMT	0.124	0.409	0.637	0.483	0.446	0.184	0.458	0.652	0.390	0.783
	AVB	0.371	0.526	0.690	0.613	0.551	0.264	0.508	0.678	0.476	0.762
	FRT	0.318	0.475	0.673	0.495	0.529	0.309	0.512	0.677	0.490	0.733
	PLD	0.611	0.624	0.623	0.648	0.615	0.398	0.542	0.601	0.518	0.635
	VNO	0.103	0.232	0.546	0.243	0.318	0.162	0.370	0.542	0.320	0.619
	WY	0.705	0.697	0.633	0.716	0.649	0.509	0.614	0.601	0.620	0.586
Utilities	AWK	0.262	0.519	0.737	0.583	0.576	0.170	0.482	0.738	0.424	0.862
	DUK	0.065	0.250	0.510	0.315	0.312	0.069	0.308	0.569	0.248	0.760
	ES	0.178	0.360	0.603	0.426	0.423	0.136	0.389	0.638	0.323	0.796
	NEE	0.170	0.343	0.578	0.371	0.397	0.125	0.362	0.609	0.297	0.767
	NRG	0.774	0.749	0.448	0.745	0.665	0.797	0.663	0.427	0.708	0.294
	PNW	0.157	0.311	0.562	0.318	0.375	0.130	0.390	0.600	0.327	0.752

Table F.14
Average selection rates of source asset subsequences for MTL-75 models of individual target assets. Source assets that belong to the same GICS sector as the respective target asset are highlighted in gray. Selection rates greater than 95% are highlighted in blue, while selection rates of less than 55% are highlighted in red. Dashes (–) indicate the absence of available subsequences for selection, due to the later distribution date of the source asset compared to the target asset.

GICS sector	Source asset	Target asset									
		TWTR	NCLH	LW	PSX	SYF	MRNA	CARR	DXC	CTVA	INVH
Communication Services	DISH	0.909	0.727	0.617	0.663	0.712	0.995	0.722	0.613	0.751	0.544
	EA	0.886	0.569	0.462	0.477	0.559	0.975	0.703	0.488	0.717	0.403
	GOOGL	0.820	0.777	0.725	0.757	0.755	0.802	0.743	0.724	0.732	0.724
	LUMN	0.651	0.813	0.833	0.865	0.833	0.780	0.759	0.821	0.774	0.825
	META	0.921	0.617	0.584	0.387	0.621	0.892	0.755	0.602	0.760	0.553
	TTWO	0.864	0.455	0.364	0.320	0.472	0.989	0.663	0.399	0.676	0.304
Consumer Discretionary	AMZN	0.908	0.746	0.611	0.687	0.723	0.905	0.688	0.621	0.695	0.588
	EBAY	0.956	0.870	0.817	0.813	0.858	0.946	0.878	0.821	0.892	0.789
	LEN	0.875	0.498	0.456	0.291	0.509	0.988	0.695	0.490	0.718	0.403
	MHK	0.945	0.766	0.677	0.673	0.749	0.961	0.772	0.684	0.806	0.621
	NWL	0.933	0.864	0.787	0.797	0.846	0.922	0.787	0.790	0.796	0.761
	TSLA	0.597	0.188	0.173	0.080	0.204	0.971	0.385	0.212	0.433	0.130
Consumer Staples	CPB	0.444	0.745	0.938	0.851	0.788	0.517	0.784	0.916	0.759	0.979
	HSY	0.399	0.711	0.911	0.846	0.747	0.429	0.740	0.887	0.704	0.962
	KMB	0.291	0.581	0.854	0.651	0.655	0.343	0.660	0.854	0.599	0.976
	PG	0.212	0.521	0.771	0.637	0.594	0.200	0.534	0.790	0.456	0.949
	TAP	0.724	0.879	0.916	0.952	0.873	0.729	0.853	0.890	0.858	0.907
	WMT	0.259	0.595	0.839	0.769	0.653	0.294	0.623	0.836	0.565	0.946
Energy	APA	0.946	0.855	0.616	0.828	0.793	0.970	0.665	0.627	0.721	0.552
	CVX	0.558	0.762	0.866	0.884	0.769	0.528	0.745	0.848	0.719	0.896
	DVN	0.976	0.865	0.603	0.849	0.801	0.985	0.677	0.615	0.727	0.547
	EQT	0.956	0.850	0.696	0.821	0.803	0.984	0.662	0.698	0.733	0.646
	HES	0.945	0.834	0.588	0.775	0.768	0.972	0.721	0.597	0.755	0.519
	XOM	0.457	0.709	0.852	0.836	0.744	0.398	0.671	0.837	0.634	0.924
Financials	AIZ	0.899	0.902	0.884	0.903	0.880	0.859	0.888	0.881	0.897	0.876
	BRK-B	0.331	0.590	0.743	0.756	0.621	0.269	0.546	0.766	0.472	0.908
	JPM	0.880	0.865	0.857	0.830	0.845	0.719	0.824	0.847	0.819	0.853
	LNC	0.931	0.759	0.713	0.667	0.757	0.989	0.812	0.726	0.845	0.676
	MTB	0.729	0.798	0.852	0.802	0.797	0.756	0.815	0.843	0.827	0.849
	NDAQ	0.944	0.865	0.852	0.818	0.851	0.777	0.853	0.840	0.851	0.839
Health Care	BIIB	0.951	0.800	0.638	0.830	0.749	0.987	0.798	0.649	0.823	0.585
	DVA	0.796	0.883	0.901	0.908	0.873	0.758	0.848	0.886	0.851	0.892
	DXCM	0.742	0.361	0.103	0.247	0.304	0.981	0.387	0.154	0.404	0.048
	JNJ	0.205	0.485	0.744	0.589	0.570	0.245	0.532	0.767	0.452	0.923
	UNH	0.904	0.881	0.844	0.849	0.870	0.771	0.842	0.837	0.833	0.842
	XRAY	0.808	0.888	0.912	0.902	0.890	0.806	0.863	0.900	0.870	0.904
Industrials	ALK	0.947	0.730	0.623	0.646	0.710	0.993	0.778	0.636	0.824	0.555
	FAST	0.953	0.897	0.861	0.846	0.888	0.942	0.906	0.856	0.911	0.837
	GNRC	0.896	0.653	0.459	0.447	0.639	0.996	0.632	0.473	0.673	0.391
	GWW	0.935	0.907	0.831	0.908	0.900	0.942	0.879	0.827	0.894	0.795
	HON	0.700	0.824	0.847	0.891	0.814	0.489	0.711	0.839	0.670	0.909
	UPS	0.442	0.691	0.841	0.865	0.721	0.424	0.694	0.831	0.642	0.922
Information Technology	AAPL	0.905	0.904	0.880	0.905	0.895	0.774	0.843	0.870	0.843	0.880
	FFIV	0.914	0.598	0.559	0.453	0.609	0.982	0.800	0.583	0.816	0.508
	FTNT	0.830	0.441	0.399	0.225	0.447	0.992	0.660	0.430	0.679	0.342
	MSFT	0.908	0.929	0.895	0.938	0.917	0.754	0.837	0.883	0.829	0.896
	QRVO	0.762	0.377	0.244	–	0.303	0.954	0.532	0.296	0.557	0.213
	TEL	0.875	0.862	0.859	0.864	0.852	0.728	0.837	0.849	0.833	0.859
Materials	APD	0.716	0.854	0.896	0.902	0.854	0.581	0.801	0.871	0.793	0.903
	IFF	0.774	0.859	0.888	0.883	0.855	0.783	0.854	0.873	0.857	0.882
	LIN	–	–	0.956	–	–	0.583	0.785	0.986	0.773	0.995
	SEE	0.933	0.830	0.798	0.789	0.825	0.950	0.864	0.804	0.871	0.773
	VMC	0.939	0.731	0.632	0.640	0.716	0.997	0.807	0.638	0.826	0.560
	WRK	0.882	0.938	0.667	–	0.541	0.992	0.795	0.680	0.831	0.615
Real Estate	AMT	0.622	0.858	0.971	0.942	0.872	0.570	0.842	0.946	0.827	0.998
	AVB	0.766	0.844	0.906	0.863	0.856	0.701	0.846	0.894	0.845	0.914
	FRT	0.761	0.836	0.897	0.873	0.852	0.780	0.839	0.887	0.855	0.896
	PLD	0.878	0.832	0.865	0.802	0.829	0.745	0.860	0.856	0.864	0.860
	VNO	0.383	0.442	0.748	0.282	0.555	0.582	0.710	0.746	0.714	0.765
	WY	0.910	0.867	0.852	0.825	0.856	0.811	0.865	0.845	0.875	0.834
Utilities	AWK	0.811	0.924	0.970	0.967	0.923	0.641	0.884	0.950	0.887	0.970
	DUK	0.297	0.639	0.896	0.659	0.755	0.336	0.675	0.881	0.630	0.986
	ES	0.517	0.794	0.934	0.893	0.818	0.449	0.749	0.905	0.720	0.976
	NEE	0.494	0.766	0.927	0.857	0.810	0.396	0.743	0.904	0.714	0.975
	NRG	0.986	0.905	0.626	0.881	0.846	0.988	0.819	0.633	0.829	0.571
	PNW	0.449	0.697	0.904	0.772	0.760	0.450	0.737	0.889	0.708	0.965

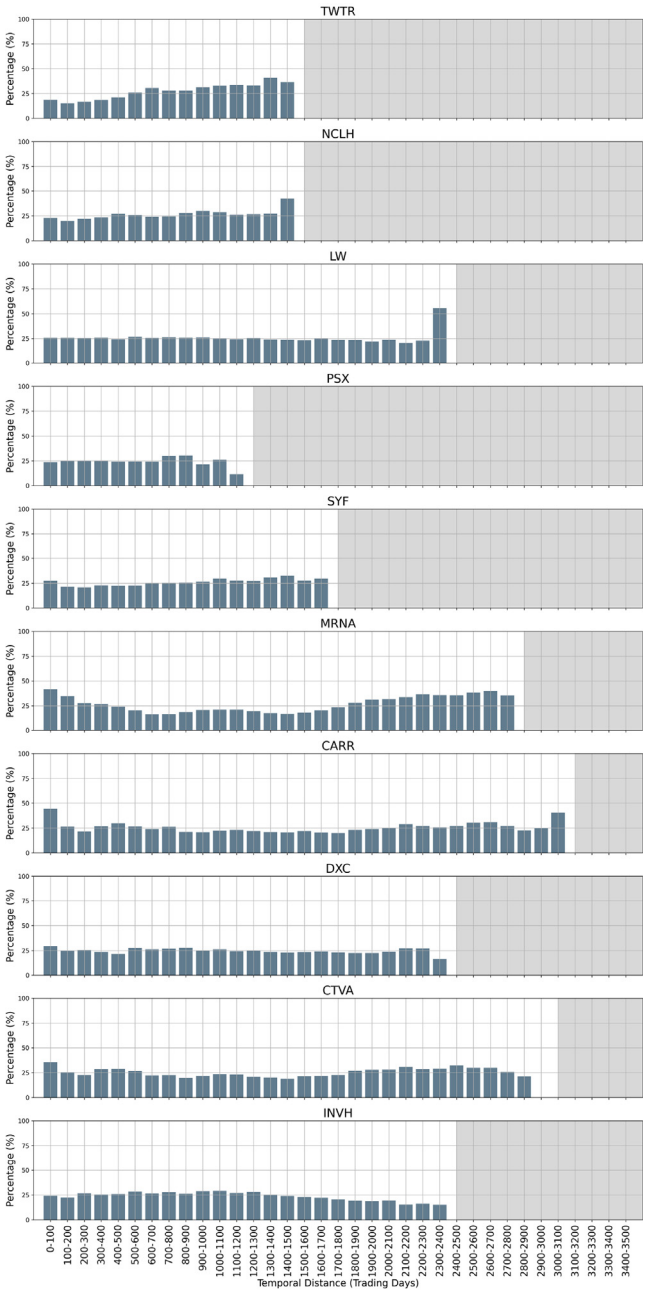


Fig. G.4. Average selection proportions of subsequences by MTL-25 models for each target asset, aggregated across all (re-)estimation steps in $s = s^*$. The selection rates are categorized based on the temporal distance between forecast origins and subsequence start dates, grouped into 100-trading-day intervals.

the boosting process:

$$\bar{Obj}_m(q) = -\frac{1}{2}\sum_{k=1}^{K_m}\frac{\left(\sum_{i=1}^n g_i I(\mathbf{x}_i \in R_{m,k})\right)^2}{\sum_{i=1}^n h_i I(\mathbf{x}_i \in R_{m,k}) + \lambda} + \gamma K_m. \quad \text{(D.8)}$$

Similar to the Gini impurity measure in classification trees or the mean squared error loss in regression trees, this criterion can be applied to evaluate potential splits during the tree-growing phase of individual weak learners. To determine splits, XGBoost in its basic implementation uses an exact greedy algorithm to evaluate all potential split candidates. This involves calculating the potential gain G of a split, based on Eq. (D.8), for each split candidate by

$$G = \frac{1}{2}\left[\frac{(\sum_{i=1}^n g_i I(\mathbf{x}_i \in R_{k_l}))^2}{\sum_{i=1}^n h_i I(\mathbf{x}_i \in R_{k_l}) + \lambda} + \frac{(\sum_{i=1}^n g_i I(\mathbf{x}_i \in R_{k_r}))^2}{\sum_{i=1}^n h_i I(\mathbf{x}_i \in R_{k_r}) + \lambda} - \frac{(\sum_{i=1}^n g_i I(\mathbf{x}_i \in R_k))^2}{\sum_{i=1}^n h_i I(\mathbf{x}_i \in R_k) + \lambda}\right] - \gamma, \quad \text{(D.9)}$$

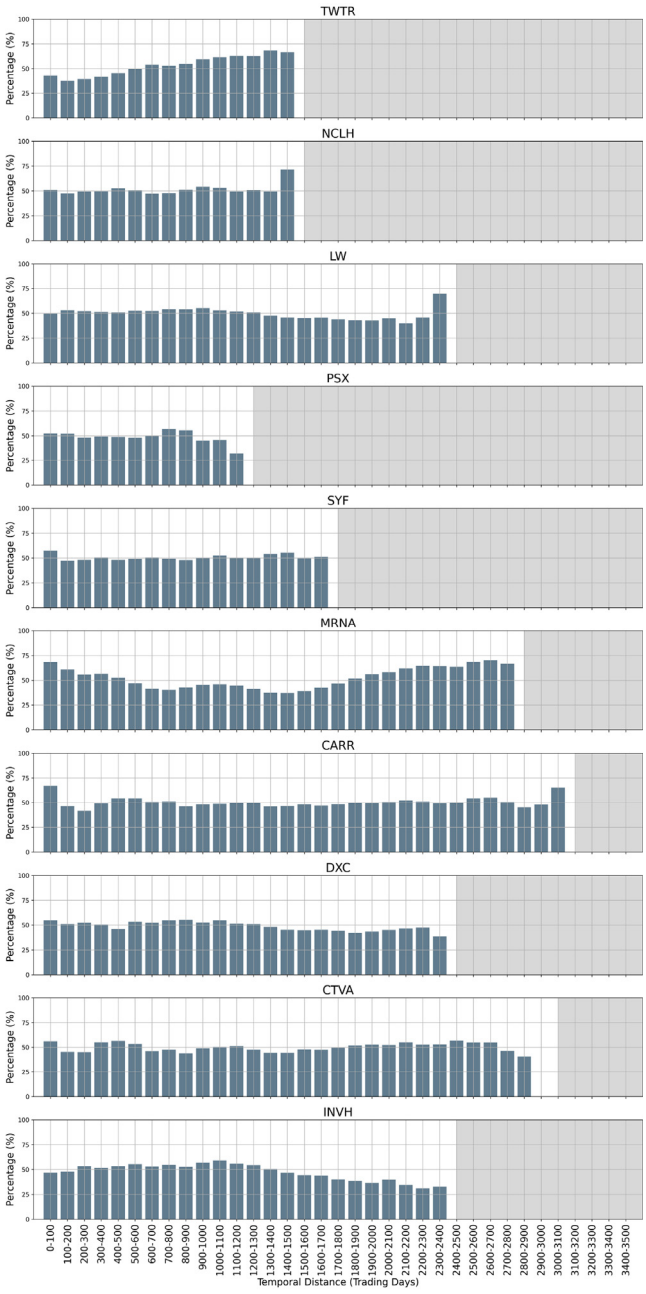


Fig. G.5. Average selection proportions of subsequences by MTL-50 models for each target asset, aggregated across all (re-)estimation steps in $s = s^*$. The selection rates are categorized based on the temporal distance between forecast origins and subsequence start dates, grouped into 100-trading-day intervals.

where $R_{k,l,}$ and $R_{k,r,}$ are the newly generated regions after adding a split. Consequently, the split that produces the highest gain G is selected in the weak learner estimation process.

Apart from the discussed L2 regularization and the regularization of the number of terminal nodes, XGBoost incorporates L1 regularization, bottom-up pruning, and top-down pruning methods, e.g. maximum tree depth restrictions or minimum observations per leaf requirements. For a detailed analysis of computational aspects such as approximate split finding or parallel tree learning, we refer the reader to Chen and Guestrin (2016).

Appendix E. Forecasting results

See Table E.10.

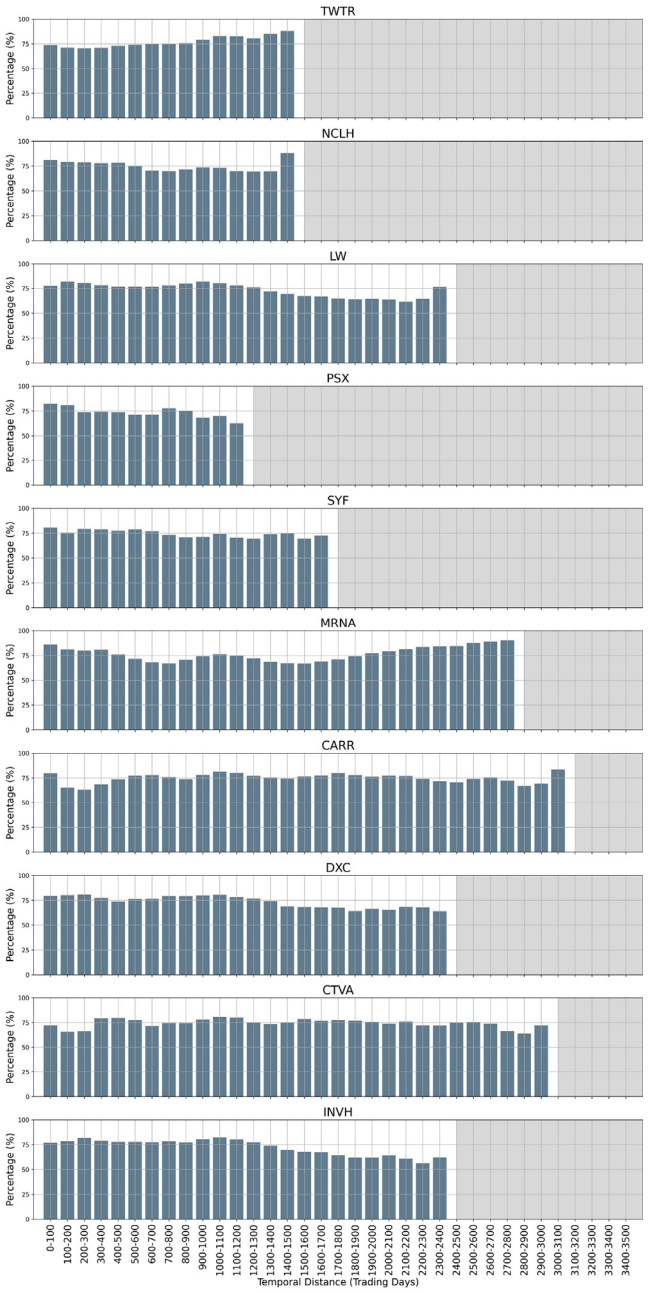


Fig. G.6. Average selection proportions of subsequences by MTL-75 models for each target asset, aggregated across all (re-)estimation steps in $s = s^*$. The selection rates are categorized based on the temporal distance between forecast origins and subsequence start dates, grouped into 100-trading-day intervals.

Appendix F. Selection rates of source assets subsequences for MTL models

See Tables F.12–F.14.

Appendix G. Temporal distances between subsequence start dates and forecast origins

See Figs. G.4–G.6.

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